

AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

October 20, 2011

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In the beginning of your answer sheet, please sign the honor code: I never received nor gave help regarding this exam.

1. (40) The Clohessy-Wiltshire equations describe the orbital perturbation dynamics of a spacecraft in a low Earth orbit, relative to the LVLH rotating coordinate frame at orbital radius $R=6,578\text{km}$. The spacecraft mass $m=100\text{kg}$. The Clohessy-Wiltshire equations can be approximated by

$$\begin{aligned}m\Delta\ddot{r} &= F_r \\mr\Delta\ddot{v} &= F_v \\mr\Delta\ddot{\lambda} &= F_\lambda\end{aligned}$$

Here $(\Delta r, \Delta v, \Delta \lambda)$ are perturbations of the spacecraft position from the circular orbit, and (F_r, F_v, F_λ) are propulsive force components expressed in the LVLH rotating coordinate frame. This approximation is often acceptable over a sufficiently small time period $T > 0$.

- a. Estimate how small the maneuver time period T should be so that the above equations are a satisfactory approximation of the Clohessy-Wiltshire equations within accuracy of about 5%.
- b. Using the above approximation, a two-impulse rendezvous maneuver is to transfer the initial spacecraft position vector $\Delta\vec{r}(0) = -10\vec{e}_r \text{ km}$ and initial velocity vector $\Delta\vec{v}(0) = 0$ to the terminal spacecraft position vector $\Delta\vec{r}(60\text{sec}) = 0$ and terminal velocity vector $\Delta\vec{v}(60\text{sec}) = 0$. What is the change in spacecraft velocity vector that occurs at each of the impulsive burns?

2. (30) A spacecraft is launched from Earth to reach the L_2 Lagrange equilibrium. At a point on its trajectory, navigation measurements indicate that the spacecraft position vector is $\vec{r} = -445,645,6\vec{i} + 500\vec{j} + 200\vec{k} \text{ km}$ and the spacecraft velocity vector is $\dot{\vec{r}} = -0.2\vec{i} - 0.3\vec{j} + 0.4\vec{k} \text{ km/sec}$, both defined with respect to Earth-Moon barycentric coordinate frame. It is desired to determine a propulsive force vector that transfers the spacecraft from this state to the L_2 Lagrange equilibrium in 7200sec.

- a. What are the linear perturbation equations that describe the spacecraft dynamics with respect to the Earth-Moon coordinate frame whose origin is located at the L_2 Lagrange point? What are the initial conditions and the terminal conditions that define the transfer maneuver?
- b. Does this transfer maneuver have a solution? That is, do you know that this transfer maneuver can be accomplished? Give your reasoning. *Comments: The problem is asking controllability*
- c. Once the spacecraft is transferred exactly to the L_2 Lagrange equilibrium, the spacecraft will always remain at this equilibrium, at least in theory. In practice, will the spacecraft always remain at this Lagrange equilibrium? Give your reasoning. *Unstable in plane out of plane*

3. (30) A 100kg spacecraft is instantaneously located at the midway point between the Earth and Moon and the spacecraft has zero velocity vector with respect to the Earth-Moon barycentric coordinate frame. Based on the restricted 3-body model, answer the following questions.

- a. If there is no propulsive force vector, is the spacecraft in equilibrium at this midway point between the Earth and Moon? If it is in equilibrium, prove your result. If it is not in equilibrium, what is the acceleration vector of the spacecraft with respect to the barycentric coordinate frame?
- b. What is the constant propulsive force vector that is required to keep the spacecraft in equilibrium at this midway point between the Earth and Moon? Express your propulsive force vector in the barycentric coordinate frame?

#1. $R = 6578 \text{ km}$ $m = 100 \text{ kg}$. $\Rightarrow W = \sqrt{\mu/R^3} \approx 1.1834 \times 10^{-3} \text{ rad/s}$.

a. The approximation eg. $\begin{cases} m \dot{r} = F_r \Rightarrow m \ddot{r} = F_r \\ r m \dot{\Delta} = F_\Delta \Rightarrow m \ddot{\Delta} = F_\Delta \\ m r \dot{\Delta} = F_\lambda \Rightarrow m \ddot{\Delta} = F_\lambda \end{cases}$

where $x = r$, $y = r \Delta$, $z = r \lambda$.

관측이 없을 때 관측자의 해는,

$$x(t) = \dot{x}_0(t) + x_0, y(t) = \dot{y}_0(t) + y_0, z(t) = \dot{z}_0(t) + z_0.$$

where $x_0, \dot{x}_0, y_0, \dot{y}_0, z_0, \dot{z}_0$ are the initial conditions.

원래의 CW의 해는 STM (state transition matrix)를 얻을 수 있는데 이를 선형화하여 근사하면 관측자의 해가 된다.

따라서 CW의 해에 있는 nonlinear term이 accuracy를 결정한다. CW의 해에 있는 비선형 항은 $\cos(\omega t)$ 과 $\sin(\omega t)$ 이다.

$$\left| \frac{\omega t - \sin(\omega t)}{\sin \omega t} \right| \leq 0.05 \quad \& \quad \left| \frac{1 - \cos(\omega t)}{\cos \omega t} \right| \leq 0.05$$

$$\Rightarrow \begin{cases} -0.05 |\sin \omega t| \leq \omega t - \sin \omega t \leq 0.05 |\sin \omega t| \\ -0.05 |\cos \omega t| \leq 1 - \cos \omega t \leq 0.05 |\cos \omega t| \end{cases}$$

$\omega t \leq \pi/2$ 라 가정하자 (ωt 가 충분히 작아야 근사를 할 수 있으므로 타당한 가정이다).

$$\Rightarrow \begin{cases} \omega t - \sin \omega t \leq 0.05 \sin \omega t \dots (B) \\ 1 - \cos \omega t \leq 0.05 \cos \omega t \dots (A) \end{cases}$$

(A)를 풀면, $(1.05)^{-1} \leq \cos \omega t \Rightarrow \omega t \leq \cos^{-1}(1.05^{-1})$

$$\Rightarrow \omega t \leq 0.3098$$

$\omega t = 0.3098$ 을 (B)에 대입하면,

$$0.3098 - 1.05 \sin(0.3098) = -0.0103 < 0.$$

따라서 0.3098은 식 (A), (B)를 모두 만족하는 수 중 가장 큰 수이다.

$$\Rightarrow t \leq 0.3098 / W \approx 261.8277 \text{ sec} \approx 4.3638 \text{ min}.$$

7% 정확도를 가지도록 하는 time period T 는 아래의 식으로 얻을 수 있다.

$$T = \frac{2\pi}{W} \approx 265.4737 \text{ sec}.$$

이는 앞에서 구한 시간 보다 크므로 사용하지 않을 것이다.

$\therefore$ The maneuver time period $T = 261.8277 \text{ sec}$.

b. 관측은 x, y, z 가 모두 분리되어 독립적이다. 초기 조건에서 x 만 0이 아니므로 y 와 z 는 고려하지 않을 것이다.

$$x(t) = \dot{x}_0 t - 10. \Rightarrow 0 = x(60\text{s}) = \dot{x}_0 \cdot 60 - 10$$

$$\Rightarrow \dot{x}_0 = 1/6.$$

$$\dot{x}(t) = \dot{x}_0 = \dot{x}_f \Rightarrow \dot{x}_f = 1/6.$$

$$\therefore \Delta V_p = 1/6 \text{ s}$$

$$\Delta V_f = -1/6 \text{ s}$$

오메가가 영인가보다!!!
->아니야

#2. $\vec{r} = -445645.6\hat{i} + 500\hat{j} + 200\hat{k}$ km. $T = 17200$ s

$\vec{v} = -0.2\hat{i} - 0.3\hat{j} + 0.4\hat{k}$ km/s

By Newton's Law, $\ddot{\vec{r}} = -\frac{\mu_E}{r_E^3} \vec{r}_E - \frac{\mu_M}{r_M^3} \vec{r}_M + \frac{\vec{F}}{m}$

where $r_E = \sqrt{(x - D_E)^2 + y^2 + z^2}$; $r_M = \sqrt{(x + D_M)^2 + y^2 + z^2}$

From $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, by Coriolis thm,

$\dot{\vec{r}} = \dot{x}\hat{i} + \dot{y}\hat{j} + \dot{z}\hat{k} + \vec{\omega} \times (x\hat{i} + y\hat{j} + z\hat{k})$

$= (\dot{x} - \omega y)\hat{i} + (\dot{y} + \omega x)\hat{j} + \dot{z}\hat{k}$

$\ddot{\vec{r}} = (\ddot{x} - \omega\dot{y})\hat{i} + (\ddot{y} + \omega\dot{x})\hat{j} + \ddot{z}\hat{k} + \dot{\vec{\omega}} \times (x\hat{i} - \omega y)\hat{i} + (\dot{y} + \omega x)\hat{j} + \dot{z}\hat{k}$

$= (\ddot{x} - 2\omega\dot{y} - \omega^2 x)\hat{i} + (\ddot{y} + 2\omega\dot{x} - \omega^2 y)\hat{j} + \ddot{z}\hat{k}$

$$\begin{cases} \ddot{x} - 2\omega\dot{y} - \omega^2 x = -\frac{\mu_E}{r_E^3} (x - D_E) - \frac{\mu_M}{r_M^3} (x + D_M) + \frac{F_x}{m} \\ \ddot{y} + 2\omega\dot{x} - \omega^2 y = -\frac{\mu_E}{r_E^3} y - \frac{\mu_M}{r_M^3} y + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E}{r_E^3} z - \frac{\mu_M}{r_M^3} z + \frac{F_z}{m} \end{cases}$$

Which is the nonlinear EoM for restric 3 body problem.

Let (x_L, y_L, z_L) be the position of L_2 point respect to the $\hat{i}\hat{j}\hat{k}$ frame.

$\Rightarrow \dot{x}_L = \dot{y}_L = \dot{z}_L = \ddot{x}_L = \ddot{y}_L = \ddot{z}_L = 0 = y_L = z_L$

Let $x = x_L + \Delta x$, $y = y_L + \Delta y = \Delta y$, $z = z_L + \Delta z = \Delta z$.

$$\frac{\mu_E}{r_E^3} = \frac{\mu_E}{[(x_L + \Delta x - D_E)^2 + \Delta y^2 + \Delta z^2]^{3/2}}$$

$$\approx \left(\frac{1}{(x_L - D_E)^2 + 2\Delta x(x_L - D_E)} \right)^{3/2} \mu_E$$

$$= \frac{\mu_E}{|x_L - D_E|^3} \left(1 + 2 \frac{\Delta x}{x_L - D_E} \right)^{-3/2} \approx \frac{\mu_E}{|x_L - D_E|^3} \left(1 - 3 \frac{\Delta x}{x_L - D_E} \right)$$

$$\frac{\mu_M}{r_M^3} = \frac{\mu_M}{[(x_L + \Delta x + D_M)^2 + \Delta y^2 + \Delta z^2]^{3/2}}$$

$$\approx \mu_M \left((x_L + D_M)^2 + 2\Delta x(x_L + D_M) \right)^{-3/2}$$

$$= \frac{\mu_M}{|x_L + D_M|^3} \left(1 + \frac{2\Delta x}{x_L + D_M} \right)^{-3/2} \approx \frac{\mu_M}{|x_L + D_M|^3} \left(1 - 3 \frac{\Delta x}{x_L + D_M} \right)$$

(i). $\ddot{\Delta x} - 2\omega\dot{\Delta y} - \omega^2 \Delta x - \cancel{\omega^2 x_L} = -W_E^2 (x_L + \Delta x - D_E) \left(1 - 3 \frac{\Delta x}{x_L - D_E} \right)$

$$- W_M^2 (x_L + \Delta x + D_M) \left(1 - 3 \frac{\Delta x}{x_L + D_M} \right) + \frac{F_x}{m}$$

$$= -W_E^2 (x_L - D_E) - W_M^2 (x_L + D_M) + 2W_E^2 \Delta x + 2W_M^2 \Delta x + F_x/m.$$

$$\Rightarrow \ddot{\Delta x} - 2\omega\dot{\Delta y} - \omega^2 (2W_E^2/\omega^2 + 2W_M^2/\omega^2 + 1) \Delta x = F_x/m.$$

$$\textcircled{a} \ddot{\Delta y} + 2W\dot{\Delta x} - W^2\Delta y = -W_E^2\Delta y \left(1 - 3\frac{\Delta x}{x_L - D_E}\right) - W_M^2\Delta y \left(1 - 3\frac{\Delta x}{x_L + D_M}\right) + \frac{F_y}{m}$$

$$\Rightarrow \ddot{\Delta y} + 2W\dot{\Delta x} + W \left(\frac{W_E^2}{W^2} + \frac{W_M^2}{W^2} - 1 \right) \Delta y = F_y/m$$

$$\textcircled{a} \ddot{\Delta z} = -W_E^2\Delta z \left(1 - 3\frac{\Delta x}{x_L - D_E}\right) - W_M^2\Delta z \left(1 - 3\frac{\Delta x}{x_L + D_M}\right) + \frac{F_z}{m}$$

$$\Rightarrow \ddot{\Delta z} + (W_E^2 + W_M^2)\Delta z = F_z/m$$

Replace $\Delta x, \Delta y, \Delta z$ to x, y, z .

$$\begin{cases} \ddot{x} - 2W\dot{y} + C_1 W^2 x = F_x/m \\ \ddot{y} + 2W\dot{x} + C_2 W^2 y = F_y/m \\ \ddot{z} + C_3 W^2 z = F_z/m \end{cases}$$

where $\begin{cases} C_1 = -2(W_E^2 + W_M^2)/W^2 - 1 \approx -17.3809 \\ C_2 = (W_E^2 + W_M^2)/W^2 - 1 \approx 2.1904 \\ C_3 = (W_E^2 + W_M^2)/W^2 \approx 3.1904 \end{cases}$

The initial state $X_0 = (x_0, \dot{x}_0, y_0, \dot{y}_0, z_0, \dot{z}_0)$
 $= (-1000, -0.2, 500, -0.3, 200, 0.4)$

The terminal state $X_f = (0, 0, 0, 0, 0, 0)$

$$\textcircled{b} \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{y} \\ \ddot{y} \\ \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -C_1 W^2 & 0 & 0 & 2W \\ 0 & 0 & 0 & 1 \\ 0 & -2W & -C_2 W^2 & 0 \\ 0 & 1 & 0 & 0 \\ -C_3 W^2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{1}{m} \end{bmatrix} \begin{bmatrix} F_x \\ F_y \end{bmatrix}$$

$$\begin{bmatrix} \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -C_3 W^2 & 0 \end{bmatrix} \begin{bmatrix} z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F_z$$

$$\Rightarrow \begin{cases} \dot{X}_{in} = A_{in} X_{in} + B_{in} U_{in} \\ \dot{X}_{out} = A_{out} X_{out} + B_{out} U_{out} \end{cases}$$

$$A_{in} B_{in} = \begin{bmatrix} \frac{1}{m} & 0 \\ 0 & \frac{2W}{m} \\ 0 & \frac{1}{m} \\ -\frac{2W}{m} & 0 \end{bmatrix} \quad A_{in}^2 B_{in} = \begin{bmatrix} 0 & \frac{2W}{m} \\ \frac{W^2}{m}(-C_1 - 4) & 0 \\ -\frac{2W}{m} & 0 \\ 0 & \frac{W^2}{m}(-4 - C_2) \end{bmatrix}$$

$$A_{in}^3 B_{in} = \begin{bmatrix} \frac{W^2}{m}(-C_1 - 4) & 0 \\ 0 & -\frac{2W^3}{m}(C_1 + C_2 + 4) \\ 0 & -\frac{W^2}{m}(C_2 + 4) \\ 2\frac{W^3}{m}(C_1 + 4 + C_2) & 0 \end{bmatrix}$$

$$\Rightarrow \text{rank} \begin{bmatrix} 0 & 0 & \frac{1}{m} & 0 & 0 & \frac{2W}{m} & -\frac{W^2}{m}(C_1 + 4) & 0 \\ \frac{1}{m} & 0 & 0 & \frac{2W}{m} & \frac{W^2}{m}(-C_1 - 4) & 0 & 0 & -\frac{2W^3}{m}(C_1 + C_2 + 4) \\ 0 & 0 & 0 & \frac{1}{m} & -\frac{2W}{m} & 0 & 0 & -\frac{W^2}{m}(C_2 + 4) \\ 0 & \frac{1}{m} & -\frac{2W}{m} & 0 & 0 & -\frac{W^2}{m}(C_2 + 4) & \frac{2W^3}{m}(C_1 + 4 + C_2) & 0 \end{bmatrix}$$

= 4.

In plane dynamics is controllable.

$$A_{out} B_{out} = \begin{bmatrix} \frac{1}{m} \\ 0 \end{bmatrix} \Rightarrow \text{rank} \begin{bmatrix} 0 & \frac{1}{m} \\ \frac{1}{m} & 0 \end{bmatrix} = 2.$$

Hence Out of plane is also controllable.

C. Consider eigenvalues of system matrix $A_{in} \Rightarrow A_{out}$.

In plane dynamics.

$$\pm \text{eigenvalues} : \pm 2.1587 \omega_0, \pm 1.8627 \omega_0 j$$

Since there exist positive real eigenvalue, the in plane dynamics is not stable.

Out of plane dynamics.

$$\pm \text{eigenvalues} : \pm 1.7862 \omega_0 j.$$

Since the eigenvalues are pure complex number, we cannot define out of plane dynamics is stable OR not.

$$\#3 \quad m = 100 \text{ kg}, \quad x_0 = (D_E - D_M) / 2 = -187699.5 \text{ km}.$$

$$\dot{x}_0 = \dot{y}_0 = \dot{z}_0 = \ddot{x}_0 = \ddot{y}_0 = \ddot{z}_0 = 0.$$

$$a. \quad \vec{F} = 0 \Rightarrow F_x = F_y = F_z = 0.$$

If the spacecraft in equilibrium, equation $\star$ in #2 holds with zero acceleration.

$$r_E = r_M = D/2 = 192374 \text{ km} = d.$$

$$\textcircled{1} \quad -\omega^2 x = + \frac{\mu_E}{d^2} - \frac{\mu_M}{d^2} = \frac{\mu_E - \mu_M}{d^2}$$

$$\text{Left Side} : -\omega^2 x \approx 1.3301 \text{e-}06.$$

$$\text{Right Side} : (\mu_E - \mu_M) / d^2 = 1.0639 \text{e-}05.$$

Since the order of both sides are different, $\textcircled{1}$ does not hold for midway point.

$\textcircled{2}$ $\textcircled{3}$ hold for midway point because both sides are zero.

Hence the spacecraft is not in equilibrium.

The acceleration $\vec{a}$ is,

$$\ddot{x} = \omega^2 x + \frac{\mu_E - \mu_M}{d^2} \approx 9.3086 \text{e-}06 \text{ km/s}^2.$$

$$\Rightarrow \vec{a} = 9.3086 \text{e-}06 \hat{i} \text{ km/s}^2.$$

b. To keep the spacecraft in equilibrium, the propulsive should be same as $-\vec{a}$.

$$\Rightarrow \frac{\vec{F}}{m} = -\vec{a} = -9.3086 \text{e-}06 \hat{i} \text{ km/s}^2.$$

AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

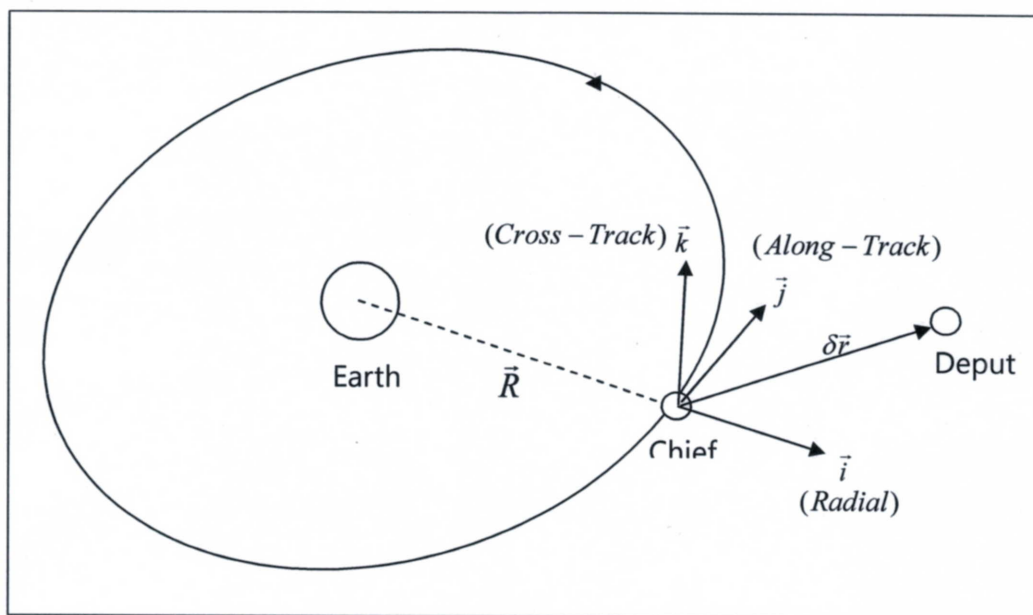
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1. (30) Consider a local-vertical-local-horizontal (LVLH) $\hat{i}\hat{j}\hat{k}$ -frame attached to a circular reference orbit whose constant orbital frequency is given by $\vec{\omega} = \omega\hat{k}$.



- a. (15) Let the position of the deputy spacecraft be given by

$$\vec{r} = \vec{R} + \delta\vec{r} = (R+x)\hat{i} + y\hat{j} + z\hat{k}.$$

Use the following Newton's Law

$$\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} + \vec{u}$$

to derive the equations of motion in the $\hat{i}\hat{j}\hat{k}$ -frame. Here μ is the gravitational parameter of the Earth, $\vec{u} = [u_x \ u_y \ u_z]^T$ is the thrust acceleration vector, and $r = \sqrt{(R+x)^2 + y^2 + z^2}$ is the distance of the deputy spacecraft from the Earth center.

- b. (15) Linearize the equations of motion derived in 1-(a) with respect to a circular reference trajectory to derive the Clohessy-Wiltshire equation.

2. (30) Suppose that a spacecraft with mass $m=100\text{kg}$ operates near a circular orbit with orbital radius $r=6,578\text{km}$. The Clohessy-Wiltshire equations can describe its orbital perturbation dynamics in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. Under some conditions, the Clohessy-Wiltshire equations can be approximated by

$$\begin{cases} m\Delta\ddot{x} = F_x \\ m\Delta\ddot{y} = F_y \\ m\Delta\ddot{z} = F_z \end{cases}$$

Here $(\Delta x, \Delta y, \Delta z)$ are perturbations of the spacecraft position from the circular orbit, and (F_x, F_y, F_z) are propulsive force components expressed in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. This approximation is often acceptable over a sufficiently small time period $T > 0$.

- (15) Estimate how small the maneuver time period T should be so that the above equations are a satisfactory approximation of the Clohessy-Wiltshire equations within accuracy of about 3%. Hint: Compare the equations of motion derived in Problem 1-(b) with the given approximation.
 - (15) Using the above approximation, a two-impulse rendezvous maneuver is to transfer the initial spacecraft position vector $\Delta\vec{r}(0) = -10\hat{i}\text{km}$ and initial velocity vector $\Delta\dot{\vec{r}}(0) = 0$ to the terminal spacecraft position vector $\Delta\vec{r}(60\text{sec}) = 0$ and terminal velocity vector $\Delta\dot{\vec{r}}(60\text{sec}) = 0$. What is the change in spacecraft velocity vector that occurs at each of the impulsive burns?
3. (25) The equations of motion for the Moon-Earth circular restricted three-body problem is given as follows:

$$\begin{cases} \ddot{x} - 2\omega_0\dot{y} - \omega_0^2x = -\frac{\mu_E(x-D_E)}{r_E^3} - \frac{\mu_M(x+D_M)}{r_M^3} + \frac{F_x}{m} \\ \ddot{y} + 2\omega_0\dot{x} - \omega_0^2y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m} \end{cases}, \begin{cases} r_E^2 = (x-D_E)^2 + y^2 + z^2 \\ r_M^2 = (x+D_M)^2 + y^2 + z^2 \\ \mu_E = 398,601\text{km}^3/\text{sec}^2 \\ \mu_M = 4,887\text{km}^3/\text{sec}^2 \\ D_E = 4,674\text{km} \\ D_M = 380,073\text{km} \\ \omega_0 = 2.66199 \times 10^{-6}\text{rad/sec} \end{cases}$$

- (10) Show that $(x_e = -444,645.6, y_e = 0, z_e = 0)$ is an equilibrium solution with no propulsive forces.
- (15) Linearize the equations of motion with respect to the equilibrium solution given in (a).

Remember

$$\omega_0^2 x_e = \frac{-\mu_E (x_e - D_E)}{(x_e - D_E)^3} - \frac{\mu_M (x_e + D_M)}{(x_e + D_M)^3}$$

4. (15) Discuss the following concepts briefly and precisely.

- (5) Halo Orbit (Original Definition in Celestial Mechanics)
- (10) Rendezvous/Orbit Transfer/Interception Maneuvers

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$$\Delta\ddot{x} - 2\omega_0\Delta\dot{y} - \omega_0^2(x_e) - \omega_0^2\Delta x$$

$$= \frac{-\mu_E}{(x_e - D_E)^2} - \frac{\mu_M}{(x_e + D_M)^2} + \frac{2\mu_E}{(x_e - D_E)^3}\Delta x + \frac{2\mu_M}{(x_e + D_M)^3}\Delta x + \frac{F_x}{m}$$

$$2 \left(\frac{\mu_E}{(x_e - D_E)^3} + \frac{\mu_M}{(x_e + D_M)^3} \right) \Delta x + \frac{F_x}{m}$$

$$\#1. \vec{r} = \vec{R} + \vec{\rho} = (R+x)\hat{i} + y\hat{j} + z\hat{k}$$

Conlon theorem,

$$\begin{aligned} \ddot{\vec{r}} &= (\ddot{R} + \dot{x})\hat{i} + \ddot{y}\hat{j} + \ddot{z}\hat{k} + (R+x)\dot{\omega}\hat{i} + y\dot{\omega}\hat{j} + z\dot{\omega}\hat{k} \\ &= \ddot{x}\hat{i} + \ddot{y}\hat{j} + \ddot{z}\hat{k} + \omega\hat{k} \times [(R+x)\hat{i} + y\hat{j} + z\hat{k}] \\ &= (\ddot{x} - \omega y)\hat{i} + (\ddot{y} + \omega(R+x))\hat{j} + \ddot{z}\hat{k} \end{aligned}$$

$$\begin{aligned} \ddot{\vec{r}} &= (\ddot{x} - \omega y)\hat{i} + (\ddot{y} + \omega(R+x))\hat{j} + \ddot{z}\hat{k} + \omega\hat{k} \times [(\ddot{x} - \omega y)\hat{i} + (\ddot{y} + \omega(R+x))\hat{j} + \ddot{z}\hat{k}] \\ &= [\ddot{x} - \omega y - \omega \ddot{y} - \omega^2(R+x)]\hat{i} + [\ddot{y} + \omega(R+x) + \omega \ddot{x} + \omega^2 y]\hat{j} + \ddot{z}\hat{k} \\ &= [\ddot{x} - 2\omega \ddot{y} - \omega^2(R+x)]\hat{i} + [\ddot{y} + 2\omega \ddot{x} - \omega^2 y]\hat{j} + \ddot{z}\hat{k} \end{aligned}$$

By Newton's Law,

$$\hat{i}: \ddot{x} - 2\omega \ddot{y} - \omega^2(R+x) = -\frac{\mu}{r^3}(R+x) + U_x$$

$$\hat{j}: \ddot{y} + 2\omega \ddot{x} - \omega^2 y = -\frac{\mu}{r^3} y + U_y$$

$$\hat{k}: \ddot{z} = -\frac{\mu}{r^3} z + U_z$$

$$\text{where } r = \sqrt{(R+x)^2 + y^2 + z^2}$$

b. For circular orbits, $\bar{x} = \bar{y} = \bar{z} = 0$.

Define $x = \bar{x} + \Delta x = \Delta x$, $y = \bar{y} + \Delta y = \Delta y$, $z = \bar{z} + \Delta z = \Delta z$.

$$\begin{aligned} \Rightarrow \frac{1}{r^3} &= \frac{1}{((R+\Delta x)^2 + \Delta y^2 + \Delta z^2)^{3/2}} = \frac{1}{(R^2 + 2\Delta x R)^{3/2}} = \frac{1}{R^3} \left(1 + 2\frac{\Delta x}{R}\right)^{-3/2} \\ &= \frac{1}{R^3} \left(1 - \frac{3\Delta x}{R}\right) \end{aligned}$$

$$\hat{i}: \Delta \ddot{x} - 2\omega \Delta \ddot{y} - \omega^2 R - \omega^2 \Delta x = -\frac{\mu}{R^3} \left(1 - \frac{3\Delta x}{R}\right) (R + \Delta x) + U_x$$

$$= -\frac{\mu}{R^3} (R - 2\Delta x) + U_x$$

$$= -\omega^2 (R - 2\Delta x) + U_x$$

$$= -\omega^2 R + 2\omega^2 \Delta x + U_x$$

$$\Rightarrow \Delta \ddot{x} - 2\omega \Delta \ddot{y} - 3\omega^2 \Delta x = U_x$$

$$\hat{j}: \Delta \ddot{y} + 2\omega \Delta \ddot{x} - \omega^2 \Delta y = -\frac{\mu}{R^3} \Delta y \left(1 - \frac{3\Delta x}{R}\right) + U_y$$

$$= -\omega^2 \Delta y + U_y$$

$$\Rightarrow \Delta \ddot{y} + 2\omega \Delta \ddot{x} = U_y$$

$$\hat{k}: \Delta \ddot{z} = -\frac{\mu}{R^3} \Delta z \left(1 - \frac{3\Delta x}{R}\right) + U_z = -\omega^2 \Delta z + U_z$$

$$\Rightarrow \Delta \ddot{z} + \omega^2 \Delta z = U_z$$

The Clohessy-Wiltshire equation

$$\begin{cases} \Delta \ddot{x} - 2\omega \Delta \ddot{y} - 3\omega^2 \Delta x = U_x \\ \Delta \ddot{y} + 2\omega \Delta \ddot{x} = U_y \\ \Delta \ddot{z} + \omega^2 \Delta z = U_z \end{cases}$$

where $\Delta x, \Delta y, \Delta z$ replaced x, y, z .

#2 a) Consider CW equation. The system matrix A is,

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ \omega^2 & 0 & 0 & +2\omega & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2\omega & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\omega^2 & 0 \end{bmatrix} \quad \text{where state } X = \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix}$$

If there is no external force, the equations of motion is $\dot{X} = AX$
 $\Rightarrow X(t) = \exp(At) X_0$ where X_0 is an initial state.

The state transition matrix $\Phi(t) = \exp(At)$ has nonlinear term for each component.

Hence the solution of CW eq. is nonlinear equation, and nonlinear terms are $\sin \omega t$ and $\cos \omega t$.

Now consider the approximation equations.

If there is no external force, $\ddot{x} = \ddot{y} = \ddot{z} = 0$.

Thus the solutions of the approximation eq are.

$$x_a(t) = \dot{x}_0 t + x_0, \quad y_a(t) = \dot{y}_0 t + y_0, \quad z_a(t) = \dot{z}_0 t + z_0$$

where $x_0, y_0, z_0, \dot{x}_0, \dot{y}_0, \dot{z}_0$ are the initial state.

Those solutions are linear approximation of the original solutions.

Hence, for any time $t < T$,

$$\left| \frac{x_a(t) - x(t)}{x(t)} \right| \leq 0.03 \quad \& \quad \left| \frac{y_a(t) - y(t)}{y(t)} \right| \leq 0.03 \quad \& \quad \left| \frac{z_a(t) - z(t)}{z(t)} \right| \leq 0.03$$

Since $\sin(\omega t)$ and $\cos(\omega t)$ are only nonlinear terms of x, y, z , then $\sin(\omega t)$ and $\cos(\omega t)$ define T .

$$\frac{\omega t - \sin(\omega t)}{\sin(\omega t)} \leq 0.03 \quad \& \quad \frac{1 - \cos(\omega t)}{\cos(\omega t)} \leq 0.03$$

$$\Rightarrow \omega t - 1.03 \sin \omega t \leq 0 \quad \& \quad 1 \leq 1.03 \cos(\omega t)$$

$$\Rightarrow \omega t \leq 0.2419$$

Since $0.2419 - 1.03 \sin(0.2419) = -4.8342 \times 10^{-3}$, then 0.2419 is the largest ωt satisfying both inequality equations.

Hence $T = 0.2419 / \omega = 0.2419 \sqrt{R^3 / \mu} \approx 204.4125 \text{ s} \approx 3.407 \text{ min}$.

Since the periods of $\sin(\omega t)$ and $\cos(\omega t)$ are $2\pi / \omega$.

Then t should be smaller than $0.03(2\pi / \omega) \approx 159.2842 \text{ sec}$.

Therefore, the maneuver time period $T = 159.2842 \text{ sec}$

b) Since all x, y, z are decoupled, then we can consider only x .

$$x(0) = -10, \quad x(60) = 0, \quad \dot{x}(0) = \dot{x}(60) = 0$$

$$\text{Since } x(t) = \dot{x}_0 t + x_0 = \dot{x}_0 t - 10, \text{ then } x(60) = 60\dot{x}_0 - 10 = 0$$

$$\Rightarrow \dot{x}_0 = 0.1667 \text{ km/s}$$

$$\text{Since } \dot{x}(t) = \dot{x}_0, \quad \dot{x}_f = \dot{x}_0 = 0.1667 \text{ km/s} \quad \Delta \vec{V}_i = 0.1667 \hat{i} \text{ km/s}$$

$$\Delta \vec{V}_f = -0.1667 \hat{i} \text{ km/s}$$

#3 If (x_e, y_e, z_e) is an equilibrium point, then all equations of motion for the M-E circular restricted 3 body problem hold for (x_e, y_e, z_e) .
 with $\dot{x}_e = \dot{y}_e = \dot{z}_e = \ddot{x}_e = \ddot{y}_e = \ddot{z}_e = 0$.

$$\textcircled{2} -\omega_0^2 x_e = -\frac{\mu_E(x_e - D_E)}{|x_e - D_E|^3} - \frac{\mu_M(x_e + D_M)}{|x_e + D_M|^3} \dots (A)$$

Left side. $-\omega_0^2 x_e = (2.66199 \times 10^{-6})^2 (-444645.6)$
 $\approx 3.1509 \times 10^{-6}$

Right side. $-\frac{\mu_E(x_e - D_E)}{|x_e - D_E|^3} = -\frac{398601(-444645.6 - 4674)}{(444645.6 + 4674)^3}$
 $= \frac{398601}{(444645.6 + 4674)^2}$
 $\approx 1.9743 \times 10^{-6}$
 $-\frac{\mu_M(x_e + D_M)}{|x_e + D_M|^3} = -\frac{4887(-444645.6 + 380073)}{(444645.6 - 380073)^3}$
 $= \frac{4887}{(444645.6 - 380073)^2}$
 $\approx 1.1717 \times 10^{-6}$

$\Rightarrow \text{Total } 3.1460 \times 10^{-6}$

Since $3.1509 \times 10^{-6} \approx 3.1460 \times 10^{-6}$, then (A) holds for (x_e, y_e, z_e) .
 The other two equations hold trivially because $y_e = z_e = 0$.
 (Both sides are zero)

Hence (x_e, y_e, z_e) is an equilibrium point.

b. Let $x = x_e + \Delta x$, $y = y_e + \Delta y$, $z = z_e + \Delta z$.

$$r_E^{-3} = [(x_e + \Delta x - D_E)^2 + \Delta y^2 + \Delta z^2]^{-3/2}$$

$$\approx [(x_e - D_E)^2 + 2\Delta x(x_e - D_E)]^{-3/2} \quad \text{HOT}$$

$$= (D_E - x_e)^3 \left[1 + 2\Delta x \frac{1}{x_e - D_E} \right]^{-3/2}$$

$$\approx (D_E - x_e)^3 \left[1 - 3\Delta x / (x_e - D_E) \right] \quad \text{Binomial thm}$$

$$r_M^{-3} = [(x_e + \Delta x + D_M)^2 + \Delta y^2 + \Delta z^2]^{-3/2}$$

$$\approx [(x_e + D_M)^2 + 2\Delta x(x_e + D_M)]^{-3/2}$$

$$= -(x_e + D_M)^3 \left[1 + 2\Delta x / (x_e + D_M) \right]^{-3/2}$$

$$\approx -(x_e + D_M)^3 \left[1 - 3\Delta x / (x_e + D_M) \right]$$

$$\textcircled{1} (\ddot{x}_e + \ddot{\Delta x}) + 2\omega_0(\dot{y}_e + \dot{\Delta y}) - \omega_0^2(x_e + \Delta x) = \frac{\mu_M}{|x_e + D_M|^3} (x_e + \Delta x + D_M) - \frac{\mu_E}{|x_e - D_E|^3} (x_e + \Delta x - D_E) + \frac{F_x}{m}$$

$$= -\frac{\mu_E}{(D_E - x_e)^3} (x_e - D_E) \left(1 - \frac{3\Delta x}{x_e - D_E} \right) + \frac{\mu_M}{(x_e + D_M)^3} (x_e + D_M) \left(1 - \frac{3\Delta x}{x_e + D_M} \right) + \frac{F_x}{m}$$

$$= -\frac{\mu_E}{(D_E - x_e)^3} (x_e - D_E) + 2\frac{\mu_E}{(D_E - x_e)^3} \Delta x + \frac{\mu_M}{(x_e + D_M)^3} (x_e + D_M) - 2\frac{\mu_M}{(x_e + D_M)^3} \Delta x + \frac{F_x}{m}$$

$$\Rightarrow \ddot{\Delta x} + 2\omega_0 \dot{\Delta y} - \omega_0^2 \Delta x = 2\frac{\mu_E}{(D_E - x_e)^3} \Delta x - 2\frac{\mu_M}{(x_e + D_M)^3} \Delta x + \frac{F_x}{m}$$

$$\Rightarrow \ddot{\Delta x} + 2\omega_0 \dot{\Delta y} - \omega_0^2 \left(1 + 2\frac{\mu_E}{\omega_0^2} - 2\frac{\mu_M}{\omega_0^2} \right) \Delta x = \frac{F_x}{m}$$

$$W_E^2 = \frac{\mu_E}{(D_E - x_e)^3} = \frac{398601}{(4674 + 444654.6)^3} \approx 0.6201 W_0^2$$

$$W_M^2 = \frac{\mu_M}{(x_e + D_M)^3} = -\frac{4887}{(444654.6 - 3800713)^3} \approx -2.5603 W_0^2$$

$$\therefore \ddot{\Delta x} - 2W_0 \dot{\Delta y} - 17.3608 W_0^2 \Delta x = \frac{F_x}{m}$$

$$\textcircled{1} \ddot{\Delta y} + 2W_0 \dot{\Delta x} - W_0^2 \Delta y = -W_E^2 \Delta y \left(1 + \frac{3\Delta x}{D_E - x_e}\right) + W_M^2 \Delta y \left(1 - \frac{3\Delta x}{x_e + D_M}\right) + \frac{F_y}{m}$$

$$= -W_E^2 \Delta y + W_M^2 \Delta y + F_y/m$$

$$\Rightarrow \ddot{\Delta y} + 2W_0 \dot{\Delta x} - (W_0^2 - W_E^2 + W_M^2) \Delta y = F_y/m$$

$$\therefore \ddot{\Delta y} + 2W_0 \dot{\Delta x} + 2.1804 W_0^2 \Delta y = \frac{F_y}{m}$$

$$\textcircled{2} \ddot{\Delta z} = -W_E^2 \Delta z \left(1 + \frac{3\Delta x}{D_E - x_e}\right) + W_M^2 \Delta z \left(1 - \frac{3\Delta x}{x_e + D_M}\right) + \frac{F_z}{m}$$

$$= -W_E^2 \Delta z + W_M^2 \Delta z + F_z/m$$

$$\Rightarrow \ddot{\Delta z} + (W_E^2 - W_M^2) \Delta z = F_z/m$$

$$\therefore \ddot{\Delta z} + 3.1804 W_0^2 \Delta z = \frac{F_z}{m}$$

Replace $\Delta x, \Delta y, \Delta z$ to x, y, z

$$\begin{cases} \ddot{x} - 2W_0 \dot{y} - 17.3608 W_0^2 x = F_x/m \\ \ddot{y} + 2W_0 \dot{x} + 2.1804 W_0^2 y = F_y/m \\ \ddot{z} + 3.1804 W_0^2 z = F_z/m \end{cases}$$

#4. a. Large, Periodic orbit such that is a solution of original nonlinear differential equations of motions

b. Rendezvous

- State to state transfer

position and velocity.

6 boundary conditions at terminal time.

- Orbit transfer

State to reduced state transfer.

5 boundary conditions at terminal time.

An angular displacement at terminal time is not considered.

- Interception

State to position transfer

3 boundary conditions at terminal time.

A velocity at terminal time is not considered.

AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

October 22, 2014 5:00-6:50 PM

Close Books/Close Notes

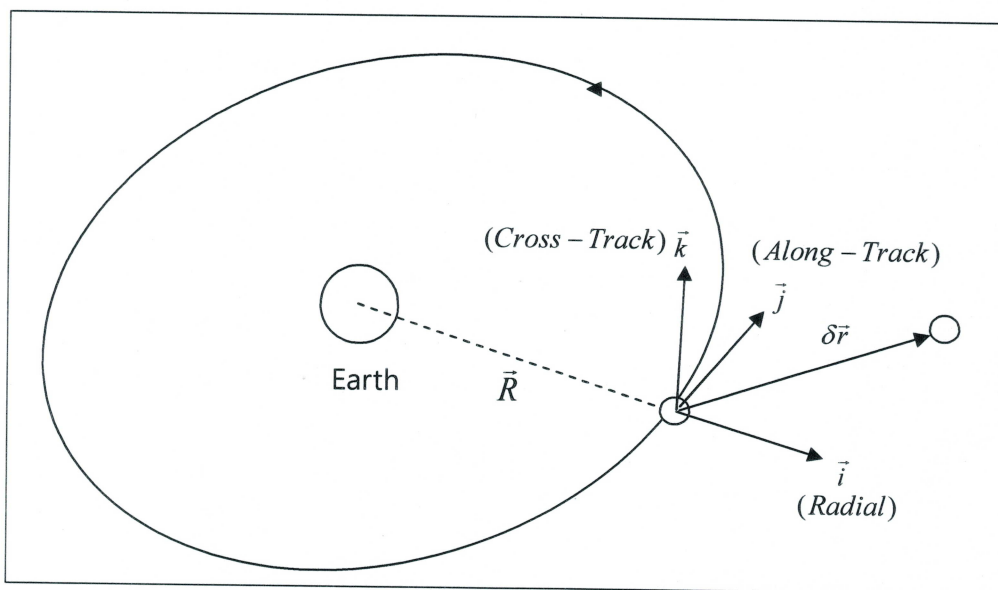
In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (20) The nonlinear differential equations that describe the dynamics of the spacecraft under the influence of the gravity of the Earth and propulsive forces are expressed in spherical coordinates as follows:

$$\begin{cases} \ddot{r} - r\dot{\nu}^2 \cos^2 \lambda - r\dot{\lambda}^2 = -\frac{\mu}{r^2} + \frac{F_r}{m} \\ r\ddot{\nu} \cos \lambda + 2\dot{r}\dot{\nu} \cos \lambda - 2r\dot{\nu}\dot{\lambda} \sin \lambda = \frac{F_\nu}{m} \\ r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\nu}^2 \sin \lambda \cos \lambda = \frac{F_\lambda}{m} \end{cases}$$

Here (r, ν, λ) represents a triple of (radial distance from the Earth center, longitude, latitude), (F_r, F_ν, F_λ) represents a triple of force components in the associated direction, m is the mass of the spacecraft, and $\mu = 4 \times 10^5 \text{ km}^3 / \text{sec}^2$ is the gravitational parameter of the Earth.

- A. (5) Suppose that the spacecraft is in a circular orbit of $(r \equiv 7,400 \text{ km}, \lambda \equiv 0)$ with no propulsive forces acting on it. What are the constant orbital velocity and orbital period of the spacecraft?
- B. (15) Linearize with respect to an arbitrary circular reference orbit to derive the Hill-Clohessy-Wiltshire equations.



2. (40) Consider a satellite initially on a geosynchronous circular equatorial orbit whose constant orbital frequency is denoted as $\vec{\omega} = \omega \hat{k}$. After performing a maneuver, it is now located at the following relative position and velocity with respect to its non-maneuvering position in the local-vertical-local-horizontal (LVLH) $\vec{i}\vec{j}\vec{k}$ -frame attached to the geosynchronous orbit:

$$\begin{cases} (x, y, z) = (-110, 70, 10) \text{ km} \\ (\dot{x}, \dot{y}, \dot{z}) = (-10, 20, 1) \text{ m/sec} \end{cases}$$

If the satellite is to initiate a two-impulse maneuver to return to its original non-maneuvering location (in 2 hours), what are the changes in velocities at the initial and terminal time, i.e., $\Delta \vec{v}_0$ and $\Delta \vec{v}_f$?

Hint: The solution to the Hill-Clohessy-Wiltshire (HCW) equation is given as

$$\begin{aligned} x(t) &= \frac{\dot{x}_0}{\omega} \sin(\omega t) - \left(3x_0 + \frac{2\dot{y}_0}{\omega} \right) \cos(\omega t) + \left(4x_0 + \frac{2\dot{y}_0}{\omega} \right) \\ y(t) &= \left(6x_0 + \frac{4\dot{y}_0}{\omega} \right) \sin(\omega t) + \frac{2\dot{x}_0}{\omega} \cos(\omega t) - (6\omega x_0 + 3\dot{y}_0)t + \left(y_0 - \frac{2\dot{x}_0}{\omega} \right) \\ z(t) &= \frac{\dot{z}_0}{\omega} \sin(\omega t) + z_0 \cos(\omega t) \end{aligned}$$

If the relative position in the HCW equation is equal to zero after time t , the initial position and velocity must satisfy the following relation:

$$\begin{cases} \dot{y}_0 = \frac{[6x_0(\omega t - \sin(\omega t)) - y_0]\omega \sin(\omega t) - 2\omega x_0(4 - 3\cos(\omega t))(1 - \cos(\omega t))}{(4\sin(\omega t) - 3\omega t)\sin(\omega t) + 4(1 - \cos(\omega t))^2} \\ \dot{x}_0 = -\frac{\omega x_0(4 - 3\cos(\omega t)) + 2(1 - \cos(\omega t))\dot{y}_0}{\sin(\omega t)} \\ \dot{z}_0 = -z_0\omega \cot(\omega t) \end{cases}$$

3. (20) A spacecraft is instantaneously located at the midway point between the Earth and Moon and the spacecraft has zero velocity vector with respect to the Earth-Moon barycentric coordinate frame. Based on the non-dimensionalized equations of motion for the restricted three-body dynamics, answer the following questions:

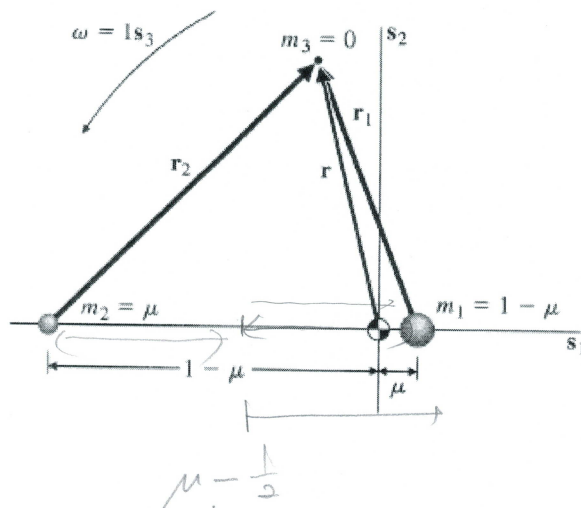


FIGURE 10.1
Geometry of the restricted problem.

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3}$$

$$\ddot{y} + 2\dot{x} - y = -\frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3}$$

$$r_1 = \sqrt{(x-\mu)^2 + y^2 + z^2}, \quad r_2 = \sqrt{(x+1-\mu)^2 + y^2 + z^2}$$

$\mu = 0.0123$ for the Earth-Moon System

$$\frac{1}{2} - \mu$$

$$\frac{3}{2} - \mu$$

- A. (15) Is the spacecraft in equilibrium at this midway point between the Earth and Moon? If yes, provide your rationale. If not, what is the acceleration vector of the spacecraft with respect to the barycentric coordinate frame?
- B. (5) What is the constant (control) acceleration vector that is required to keep the spacecraft in equilibrium at this midway point between the Earth and Moon? Express your control acceleration vector in the barycentric coordinate frame.
4. (20) Consider a two-dimensional linear system

$$\Delta \dot{x} = A \Delta x, \quad \Delta x(0) = \Delta x_0$$

Let $\lambda = \pm j\omega$ and $(\xi, \bar{\xi})$ be the eigenvalues and eigenvectors of the system, respectively. Describe how to find the initial conditions that compose a Halo Orbit with a single frequency of ω .

$$\text{span}\{\text{Re}\xi, \text{Im}\xi\}$$

$$e^{At} x_0$$

$$\begin{bmatrix} e^{At} & e^{-At} \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

$$x(t) = e^{At} x_0$$

$$\mu = 4e+05 \text{ km}^3/\text{sec}^2$$

$$\ddot{r} - r\dot{\lambda}^2 \cos^2 \lambda - r\dot{\lambda}^2 = -\mu/r^2 + F_r/m \dots \textcircled{a}$$

$$r\ddot{\lambda} \cos \lambda + 2\dot{r}\dot{\lambda} \cos \lambda - 2r\dot{\lambda}\dot{\lambda} \sin \lambda = F_\lambda/m \dots \textcircled{b}$$

$$r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\lambda}^2 \sin \lambda \cos \lambda = F_\lambda/m \dots \textcircled{c}$$

$$r = 17400 \text{ km}, \lambda = 0, \vec{F} = 0 \Rightarrow F_r = F_\lambda = F_\lambda = 0$$

$$\text{Since } r = 17400 \text{ km} \Rightarrow \dot{r} = 0 \text{ km/s} \text{ \& } \ddot{r} = 0 \text{ km/s}^2$$

$$\text{Since } \lambda = 0, \dot{\lambda} = 0 \text{ rad/s} \text{ \& } \ddot{\lambda} = 0 \text{ rad/s}^2$$

$$\Rightarrow (17400 \text{ km})\ddot{\lambda} = +\mu/(17400 \text{ km})^2 \text{ from } \textcircled{a}$$

$$(17400 \text{ km})\ddot{\lambda} = 0 \text{ from } \textcircled{b}$$

The both sides of $\textcircled{c}$ are zero.

$$\Rightarrow \ddot{\lambda} = +\mu/(17400 \text{ km})^3 \text{ \& } \ddot{\lambda} = 0$$

$$\Rightarrow \dot{\lambda} = \sqrt{\frac{4e+05 \text{ km}^3/\text{sec}^2}{(17400 \text{ km})^3}} \approx 9.935 \text{ e-04 rad/s}$$

$$\text{Thus, the constant orbital velocity } \vec{v} = r\dot{\lambda} = 7.352 \text{ km/s}$$

$$\text{The orbital period is } T = 2\pi/\dot{\lambda} \approx 6324.293 \text{ s} = 1.757 \text{ hours.}$$

B. Let R be a radius of an arbitrary circular reference orbit

$$\text{Since the orbit is circular, } \lambda^* = 0 = \dot{\lambda}^* = \ddot{\lambda}^* \text{ and } \dot{R} = 0 = \ddot{R}$$

$$\text{From } \textcircled{a}, -R\dot{\lambda}^2 = -\mu/R^2 \Rightarrow \dot{\lambda} = \sqrt{\mu/R^3} \equiv \omega_0$$

$$\text{From } \textcircled{b}, R\ddot{\lambda} = 0 \Rightarrow \ddot{\lambda} = 0$$

The both sides of $\textcircled{c}$ are zero.

$$\text{Hence } \dot{\lambda}(t) = \omega_0 t + C \text{ for some constant } C.$$

$$\text{Define } r = R + \Delta r, \lambda = 0 + \Delta \lambda, \text{ and } \dot{\lambda} = \dot{\lambda}^* + \Delta \dot{\lambda} \text{ for equilibrium solutions } (R, \dot{\lambda}^*, 0).$$

From $\textcircled{a}$,

$$\begin{aligned} \text{Left side: } & (\ddot{R} + \ddot{\Delta r}) - (R + \Delta r)(\dot{\lambda}^* + \Delta \dot{\lambda})^2 \cos^2 \Delta \lambda - (R + \Delta r)\ddot{\Delta \lambda}^2 \\ & = \ddot{\Delta r} - (R + \Delta r)(\dot{\lambda}^{*2} + 2\Delta \dot{\lambda}\dot{\lambda}^*) \\ & = \ddot{\Delta r} - R\omega_0^2 - \omega_0^2 \Delta r - 2R\omega_0 \Delta \dot{\lambda} \end{aligned}$$

$$\begin{aligned} \text{Right side: } & -\mu/(R + \Delta r)^2 + F_r/m \\ & = -(\mu/R^2)(1 + 2\frac{\Delta r}{R})^{-1} + F_r/m \\ & = -(\mu/R^2)(1 - 2\frac{\Delta r}{R}) \\ & = -\omega_0^2(R - 2\Delta r) + F_r/m \\ & = -\omega_0^2 R + 2\omega_0^2 \Delta r + F_r/m \end{aligned}$$

$$\Rightarrow \ddot{\Delta r} - 2R\omega_0 \Delta \dot{\lambda} - \omega_0^2 \Delta r - R\omega_0^2 = -\omega_0^2 R + 2\omega_0^2 \Delta r + F_r/m$$

$$\Rightarrow \ddot{\Delta r} - 2R\omega_0 \Delta \dot{\lambda} - 3\omega_0^2 \Delta r = F_r/m \dots \textcircled{d}$$

From $\textcircled{b}$

$$\begin{aligned} \text{Left side: } & (R + \Delta r)(\ddot{\lambda}^* + \ddot{\Delta \lambda}) \cos \Delta \lambda + 2(R + \Delta r)(\dot{\lambda}^* + \Delta \dot{\lambda}) \dot{\Delta \lambda} \sin \Delta \lambda - 2(R + \Delta r)\dot{\lambda}^* \dot{\Delta \lambda} \\ & = (R + \Delta r)\ddot{\Delta \lambda} + 2\dot{\Delta r}(\omega_0 + \Delta \dot{\lambda}) \\ & = R\ddot{\Delta \lambda} + 2\omega_0 \dot{\Delta r} \end{aligned}$$

$$\Rightarrow R\ddot{\Delta \lambda} + 2\omega_0 \dot{\Delta r} = F_\lambda/m \dots \textcircled{e}$$

From (C),

$$\begin{aligned} \text{Left side : } (R+\Delta r) \ddot{\Delta \lambda} + 2(R+\Delta r) \dot{\Delta \lambda} \dot{\Delta \lambda}^0 + (R+\Delta r) (\dot{\Delta \lambda}^0 + \Delta \mu)^2 \sin \Delta \lambda \cos \Delta \lambda^1 \\ = R \ddot{\Delta \lambda} + (R+\Delta r) (\omega_0^2 + 2\omega_0 \Delta \mu) \Delta \lambda \\ = R \ddot{\Delta \lambda} + R \omega_0^2 \Delta \lambda \end{aligned}$$

$$\Rightarrow R \ddot{\Delta \lambda} + R \omega_0^2 \Delta \lambda = F_r / m \dots \textcircled{4}$$

Define $x = \Delta r$, $y = R \Delta \mu$ and $z = R \Delta \lambda$.

$$\textcircled{4} : \ddot{x} - 2\omega_0 \dot{y} - \omega_0 x = F_r / m \dots \textcircled{4}'$$

$$\textcircled{5} : \ddot{y} - 2\omega_0 \dot{x} = F_y / m \dots \textcircled{5}'$$

$$\textcircled{6} : \ddot{z} + \omega_0 z = F_z / m \dots \textcircled{6}'$$

Those equations are called Hill-Clohesy-Wiltshire equations.

$$\#2. [x, y, z]^T = [-110, 70, 10]^T \text{ km.}$$

$$[\dot{x}, \dot{y}, \dot{z}]^T = [-10, 20, 1]^T \text{ km/s.}$$

$$\text{Maneuvering time } T = 2h = 7200 \text{ s.}$$

$$\text{Initial condition } [x_0, \dot{x}_0, y_0, \dot{y}_0, z_0, \dot{z}_0]^T = [-110, -0.01, 70, 0.02, 10, 0.001]^T$$

$$\text{Final condition } [x_f, \dot{x}_f, y_f, \dot{y}_f, z_f, \dot{z}_f]^T = [0, 0, 0, 0, 0, 0]^T$$

$R = 38500 \text{ km}$: the radius of the reference frame.

$$\omega = +\sqrt{\frac{\mu}{R^3}} = \sqrt{\frac{4E+05 \text{ km}^3/\text{s}^2}{(38500 \text{ km})^3}} \approx 8.372E-05 \text{ rad/s.}$$

By the hint, $\omega T \approx 0.601$

$$\dot{y}_0^+ = \frac{[6x_0(\omega T - \sin \omega T) - y_0] \omega \sin(\omega T) - 2\omega x_0(4 - 3\cos \omega T)(1 - \cos \omega T)}{(4\sin \omega T - 3\omega T)\sin \omega T + 4(1 - \cos \omega T)^2}$$

$$\Rightarrow \text{Numerator : } 6x_0(\omega T - \sin \omega T) = -660(0.601 - \sin(0.601)) \approx -23.451$$

$$\omega \sin(\omega T) = 8.372E-05 \times \sin(0.601) \approx 4.734E-05$$

$$2\omega x_0(4 - 3\cos \omega T) = 2 \times 8.372E-05 \times (-110) \times (4 - 3\cos(0.601)) \approx -0.0281$$

$$1 - \cos \omega T = 1 - \cos(0.601) \approx 0.175$$

$$\Rightarrow \text{Numerator is } -4.424E-03 + 4.918E-03 = 4.935E-04$$

$$\Rightarrow \text{Denominator : } (4\sin \omega T - 3\omega T)\sin \omega T$$

$$= (4\sin(0.601) - 3 \times 0.601)\sin(0.601)$$

$$\approx 0.259$$

$$4(1 - \cos \omega T)^2 = 4(1 - \cos(0.601))^2$$

$$\approx 0.1228$$

$$\Rightarrow \text{Denominator is } 0.3818$$

$$\therefore \dot{y}_0^+ = +1.293E-03 \text{ km/s.}$$

$$\dot{x}_0^+ = -\frac{\omega x_0(4 - 3\cos(\omega T)) + 2(1 - \cos(\omega T)) \dot{y}_0}{\sin(\omega T)}$$

$$\Rightarrow \text{Numerator : } \omega x_0(4 - 3\cos(\omega T)) = 8.372E-05 \times (-110) \times (4 - 3\cos(0.601)) \approx -0.0141$$

$$2(1 - \cos(\omega T)) \dot{y}_0 = 2(1 - \cos(0.601)) (1.243 \times 10^{-3}) \\ \approx 4.531 \times 10^{-4}$$

⇒ Numerator is -0.0136

⇒ Denominator: $\sin \omega T = \sin(0.601) \approx 0.565$

∴ $\dot{x}_0^+ \approx 0.0242 \text{ km/s}$

$\dot{z}_0^+ = -z_0 \omega \cot(\omega T) = -10 \times 3.872 \times 10^{-5} \times \cot(0.601) \\ \approx -0.0306 \text{ km/s}$

⇒ $\Delta \vec{V}_0 = (\dot{x}_0^+ - \dot{x}_0) \vec{e}_r + (\dot{y}_0^+ - \dot{y}_0) \vec{e}_\mu + (\dot{z}_0^+ - \dot{z}_0) \vec{e}_\lambda \\ = (0.0242 + 0.01) \vec{e}_r + (1.243 \times 10^{-3} - 0.02) \vec{e}_\mu + (-0.0306 - 0.001) \vec{e}_\lambda \\ = [0.0342, -0.0187, -0.0316] \text{ km/s}$

⇒ $\Delta V_0 = \|\Delta \vec{V}_0\| = 0.0502 \text{ km/s}$

From Hints

$x(t) = \frac{\dot{x}_0^+}{\omega} \sin(\omega t) - (3x_0 + \frac{2\dot{y}_0^+}{\omega}) \cos(\omega t) + (4x_0 + \frac{2\dot{y}_0^+}{\omega})$

⇒ $\dot{x}(t) = \dot{x}_0^+ \cos(\omega t) + (3x_0 \omega + 2\dot{y}_0^+) \omega \sin(\omega t)$

⇒ $\dot{x}_F = \dot{x}(T) = \dot{x}_0^+ \cos(\omega T) + (3x_0 \omega + 2\dot{y}_0^+) \omega \sin(\omega T) \\ = 0.0242 \cos(0.601) + (3 \times 0.0242 \times 3.872 \times 10^{-5} + 2 \times 1.243 \times 10^{-3}) \omega \sin(0.601) \\ \approx 0.0200 + 2.592 \times 10^{-3} \times 4.734 \times 10^{-3} \\ = 0.0200 \text{ km/s}$

$y(t) = (6x_0 + \frac{4\dot{y}_0^+}{\omega}) \sin(\omega t) + \frac{2x_0}{\omega} \cos(\omega t) - (6\omega x_0 + 3\dot{y}_0^+)t + (y_0 - \frac{2x_0}{\omega})$

⇒ $\dot{y}(t) = (6\omega x_0 + 4\dot{y}_0^+) \cos(\omega t) - 2x_0 \sin(\omega t) - (6\omega x_0 + 3\dot{y}_0^+)$

⇒ $\dot{y}_F = \dot{y}(T) = (6\omega x_0 + 4\dot{y}_0^+) \cos(\omega T) - 2x_0 \sin(\omega T) - (6\omega x_0 + 3\dot{y}_0^+) \\ = (6 \times 3.872 \times 10^{-5} \times (-110) + 4 \times (1.243 \times 10^{-3})) \cos(0.601) \\ - 2 \times (0.0242) \sin(0.601) - (6 \times 3.872 \times 10^{-5} \times (-110) + 3 \times (1.243 \times 10^{-3})) \\ \approx -0.0168 - 0.0274 + 0.0217 \\ = -0.0225 \text{ km/s}$

$z(t) = \frac{\dot{z}_0^+}{\omega} \sin(\omega t) + z_0 \cos(\omega t)$

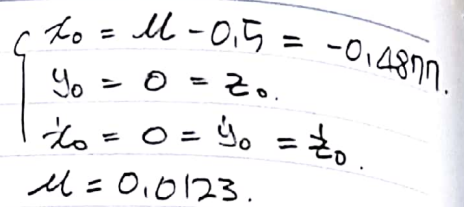
⇒ $\dot{z}(t) = \dot{z}_0^+ \cos(\omega t) - z_0 \omega \sin(\omega t)$

⇒ $\dot{z}_F = \dot{z}(T) = \dot{z}_0^+ \cos(\omega T) - z_0 \omega \sin(\omega T) \\ = -0.0306 \cos(0.601) - 10 \times 3.872 \times 10^{-5} \times \sin(0.601) \\ \approx -0.0252 - 2.189 \times 10^{-4} \\ = -0.0254 \text{ km/s}$

$\Delta \vec{V}_F = [-0.0200, 0.0225, 0.0254] \text{ km/s}$

⇒ $\Delta V_F = 0.0394 \text{ km/s}$

#3.



3 body system holds for the state of the spacecraft. ($\ddot{x} = \ddot{y} = \ddot{z} = 0$).

$$\Rightarrow 0 - 2 \times 0 + 0,48777 = - \frac{(1 - 0,0123)(-0,48777 - 0,0123)}{(0,48777 + 0,0123)^3} - \frac{0,0123(-0,48777 - 0,0123)}{(-0,48777 + 1 - 0,0123)^3}$$

$\Rightarrow$ left side : 0.4877.

Right side : $3.9508 - 0.1449 = 3.8059$

The left side and right side are totally different.

Hence the spacecraft is not in equilibrium.

The second and third equations hold because both sides are zero.

Thus there exist only $\hat{z}$ -axis component acceleration on the spacecraft w.r.t the barycenter frame.

The acceleration is,

$$\frac{11}{x} = 2y + x - \frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3} \approx 3.8059 - 0.48177 = 3.3182$$

B. For keeping the spacecraft in equilibrium at this midway points make the first equations of motion hold.

The other equations of motion are already hold, then we consider only $\hat{t}$ -axis term.

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3} + \frac{F_x}{m}.$$

$$\Rightarrow 0.4877 = 3.80594 F_x / m$$

$$\Rightarrow F_x/m = 0,4877 - 3,8059 \approx -3,3182.$$

Hence, the control acceleration is $-3.3182 \hat{e}$.

#4. $\Delta x \in \mathbb{R}^2$. $\Delta'x = A\Delta x$, $\Delta x(0) = \Delta x_0$. Let λ be eigenvalues of A .

Let $\lambda = \pm j\omega$ and $\xi, \bar{\xi}$ be eigenvectors corresponding to $j\omega, -j\omega$.

The solution of $\dot{x} = Ax$ is,

$$\Delta x = e^{At} \Delta x_0 = a_1 \frac{1}{\sqrt{2}} e^{-\sqrt{2}t} + a_2 \frac{1}{\sqrt{2}} e^{-\sqrt{2}t} \text{ for some } a_1, a_2.$$

$$= a_1 \xi (\cos \omega t + j \sin \omega t) + a_2 \xi (\cos \omega t - j \sin \omega t)$$

$$= (a_1 \xi + a_2 \bar{\xi}) \cos \omega t + j (a_1 \xi - a_2 \bar{\xi}) \sin \omega t$$

Since $\Delta\pi(0) = \Delta\pi_0$, $\Delta\pi_0 = a_1\tilde{\xi} + a_2\tilde{\eta}$.

Also, $\delta \dot{x}_0 = j\omega(a_1 \xi_1 - a_2 \bar{\xi}_1)$

$$\Delta x_0 = [\xi \mid \bar{\xi}] \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$\Rightarrow \Delta \dot{x}_0 = j\omega [\xi \mid -\bar{\xi}] \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = [\xi \mid \bar{\xi}] \begin{bmatrix} j\omega & 0 \\ 0 & -j\omega \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}.$$

Solve for (a_1, a_2) .

for Δx_0 composing a Halo Orbit, $\Delta x_0 \in \mathbb{R}^2$.

for $\Delta x_0 \in \text{Span}\{\text{Im}\xi, \text{Re}\xi\}$

It means, if $a_1 = a_2$, $\Delta x_0 = 2a_1 \text{Re}\xi$.

for example, if $a_1 = -a_2$, $\Delta x_0 = 2a_2 \text{Im}\xi$.

AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

October 21, 2015 1:00-2:50 PM

Close Books/Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (30) Suppose that a spacecraft with mass $m=100\text{kg}$ operates near a circular orbit with orbital radius $r=6,578\text{km}$. The Clohessy-Wiltshire equations can describe its orbital perturbation dynamics in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. Under some conditions, the Clohessy-Wiltshire equations can be further approximated by

$$m\Delta\ddot{x} = F_x$$

$$m\Delta\ddot{y} = F_y$$

$$m\Delta\ddot{z} = F_z$$

$$\begin{aligned} 100\Delta\ddot{x} &= F_x \\ 100\Delta\ddot{y} &= F_y \\ 100\Delta\ddot{z} &= F_z \end{aligned}$$

Here $(\Delta x, \Delta y, \Delta z)$ are perturbations of the spacecraft position from the circular orbit, and (F_x, F_y, F_z) are propulsive force components expressed in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. This approximation is often acceptable over a sufficiently small time period $T > 0$.

- a. (15) Estimate how small the maneuver time period T should be so that the above equations are a satisfactory approximation of the Clohessy-Wiltshire equations within accuracy of about 3%. Hint: Consider the following analytic solutions to the Clohessy-Wiltshire equations:

$$x(t) = \frac{\dot{x}_0}{\omega} \sin(\omega t) - \left(3x_0 + \frac{2\dot{y}_0}{\omega}\right) \cos(\omega t) + \left(4x_0 + \frac{2\dot{y}_0}{\omega}\right)$$

$$y(t) = \left(6x_0 + \frac{4\dot{y}_0}{\omega}\right) \sin(\omega t) + \frac{2\dot{x}_0}{\omega} \cos(\omega t) - (6\omega x_0 + 3\dot{y}_0)t + \left(y_0 - \frac{2\dot{x}_0}{\omega}\right)$$

$$z(t) = \frac{\dot{z}_0}{\omega} \sin(\omega t) + z_0 \cos(\omega t)$$

$$T = \frac{2\pi}{\omega}$$

$$0.03T = 0.03 \frac{2\pi}{\omega}$$

- b. (15) Using the above approximation, a two-impulse rendezvous maneuver is to transfer the initial spacecraft position vector $\Delta\vec{r}(0) = -10\hat{i}\text{km}$ and initial velocity vector $\Delta\vec{v}(0) = 0$ to the terminal spacecraft position vector $\Delta\vec{r}(60\text{sec}) = 0$ and terminal velocity vector $\Delta\vec{v}(60\text{sec}) = 0$. What is the change in spacecraft velocity vector that occurs at each of the impulsive burns?

$$\omega = \sqrt{\frac{\mu}{R^3}}$$

$$T = \frac{2\pi}{\omega}$$

$$\omega^2 x = F_x$$

$$\omega \rightarrow 1 \quad T = \frac{2\pi}{\omega}$$

$$\frac{6\pi}{100}$$

$$C_0 \dot{y}_0 = \frac{3\mu}{2\pi\omega R^2}$$

$$v = 0.10$$

$$\Delta \vec{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

2. (40) Consider the equations of motion for the Earth-Moon circular restricted three-body problem:

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_E (x - D_E)}{r_E^3} - \frac{\mu_M (x + D_M)}{r_M^3} + \frac{F_x}{m} \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m} \end{cases} \quad , \quad \begin{cases} r_E^2 = (x - D_E)^2 + y^2 + z^2 \\ r_M^2 = (x + D_M)^2 + y^2 + z^2 \\ \mu_E = 398,601 \text{ km}^3 / \text{sec}^2 \\ \mu_M = 4,887 \text{ km}^3 / \text{sec}^2 \\ D_E = 4,674 \text{ km} \\ D_M = 380,073 \text{ km} \\ \omega_0 = 2.66199 \times 10^{-6} \text{ rad/sec} \end{cases}$$

- (5) Show that $L_4 : (\bar{x} = -187,699.3 \text{ km}, \bar{y} = 333,201.5 \text{ km}, \bar{z} = 0)$ is a Lagrangian point.
- (20) Linearize the above equations of motion with respect to the Lagrangian point given in (a).
- (5) What are the eigenvalues of the linearized out-of-plane equations of motion?
- (10) Discuss the stability of out-of-plane motion based on its eigenvalues.

3. (20) For the 3-2-1 rotation, derive the angular velocity components $\vec{\omega} = [\omega_x, \omega_y, \omega_z]^T$ in terms of the Euler angles (ψ, θ, ϕ) and their rates.

4. (10) Discuss the following concepts briefly and precisely.

- (5) Halo Orbit (Original Definition in Celestial Mechanics)

- (5) Exponential Coordinates

enc

Δ

μ

$$\#1. R = 6578 \text{ km}, m = 100 \text{ kg}.$$

The original CW equations.

$$\begin{cases} \ddot{\Delta x} - 2\omega_0 \dot{\Delta y} + \omega_0^2 \Delta x = F_r/m \\ \ddot{\Delta y} - 2\omega_0 \dot{\Delta x} = F_y/m \\ \ddot{\Delta z} + \omega_0^2 \Delta z = F_z/m \end{cases}$$

a. Since $R = 6578 \text{ km}$, a distance from the origin of the reference frame to a location of the spacecraft has to be less than or equal to $0.03R \approx 197.34 \text{ km}$.

From $m\ddot{\Delta x} = F_r$, $m\ddot{\Delta y} = F_y$, and $m\ddot{\Delta z} = F_z$,

$$\begin{cases} \Delta x(t) = \dot{x}_0 t + x_0 \\ \Delta y(t) = \dot{y}_0 t + y_0 \\ \Delta z(t) = \dot{z}_0 t + z_0 \end{cases}$$

Those 3 equations are linear approximation of the 3 equations of the hint. Consider $x(t)$ of the hint

Nonlinear terms are $\sin(\omega t)$ and $\cos(\omega t)$, and ωt and 1 is approximated terms of $\sin \omega t$ and $\cos \omega t$.

Since required accuracy is 3%,

$$\omega t - \sin(\omega t) \leq 0.03 \sin(\omega t) \quad \& \quad 1 - \cos(\omega t) \leq 0.03 \cos(\omega t)$$

$$\Rightarrow \omega t \leq 1.03 \sin(\omega t) \quad \& \quad 1 \leq 1.03 \cos(\omega t)$$

$$\Rightarrow \omega t \leq 0.4200 \quad \& \quad \omega t \leq 0.2419.$$

$$\Rightarrow \omega t \leq 0.2419.$$

$$\therefore t \leq \frac{0.2419}{\omega} = 0.2419 \int \frac{R^3}{\mu} \approx 204.4128 \text{ sec} \approx 3.4061 \text{ min.}$$

b. The initial state $X_0 = [-10, 0, 0, 0, 0, 0]^T$

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} X_0^+ = A X_0^+ \quad \text{where} \quad X_0^+ = \begin{bmatrix} -10 \\ \dot{x}_0^+ \\ 0 \\ \dot{y}_0^+ \\ 0 \\ \dot{z}_0^+ \end{bmatrix}$$

$$\Rightarrow X(t) = e^{At} X_0^+ = \left[\sum_{i=0}^{\infty} \frac{(At)^i}{i!} \right] X_0^+ = (I + At) X_0^+$$

$$\Rightarrow \begin{bmatrix} 0 \\ \dot{x}_f \\ 0 \\ \dot{y}_f \\ 0 \\ \dot{z}_f \end{bmatrix} = \begin{bmatrix} 1 & T & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & T & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & T \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -10 \\ \dot{x}_0^+ \\ 0 \\ \dot{y}_0^+ \\ 0 \\ \dot{z}_0^+ \end{bmatrix}$$

$$\Rightarrow \begin{cases} -10 + \dot{x}_0^+ T = 0 \\ 0 + \dot{y}_0^+ T = 0 \\ 0 + \dot{z}_0^+ T = 0 \end{cases} \quad \& \quad \begin{cases} \dot{x}_0^+ = \dot{x}_f \\ \dot{y}_0^+ = \dot{y}_f \\ \dot{z}_0^+ = \dot{z}_f \end{cases}$$

$$\Rightarrow \dot{x}_0^+ = 1.667 \text{ km/s} = \dot{x}_f, \quad 0 = \dot{y}_0^+ = \dot{z}_0^+ = \dot{y}_f = \dot{z}_f$$

Thus $\Delta \vec{V}_0 = 1.667 \hat{i} \text{ km/s}$ and $\Delta \vec{V}_f = -1.667 \hat{i} \text{ km/s}$

#2. EoM for the Earth-Moon circular restricted 3-body problem.

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_e(x - D_e)}{r_e^3} - \frac{\mu_m(x + D_m)}{r_m^3} + \frac{F_x}{m} \quad (a) \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_e y}{r_e^3} - \frac{\mu_m y}{r_m^3} + \frac{F_y}{m} \quad (b) \\ \ddot{z} = -\frac{\mu_e z}{r_e^3} - \frac{\mu_m z}{r_m^3} + \frac{F_z}{m} \quad (c) \end{cases}$$

a. If $L_4: (\bar{x}, \bar{y}, \bar{z}) = (-187699.3, 333201.5, 0)$ km is a equilibrium point, 3 EoM hold for L_4 with zero velocity and zero acceleration.

$$(a): -\omega_0^2 \bar{x} = -\frac{\mu_e(\bar{x} - D_e)}{r_e^3} - \frac{\mu_m(\bar{x} + D_m)}{r_m^3} + \frac{F_x}{m} \quad (a)$$

$$\text{Left side: } -\omega_0^2 \bar{x} = (2.66199 \times 10^{-6})^2 (-187699.3) \approx -1.3301 \times 10^{-6}$$

$$\text{Right side: } -\frac{\mu_e(\bar{x} - D_e)}{r_e^3} - \frac{\mu_m(\bar{x} + D_m)}{r_m^3} \quad (d)$$

$$\begin{aligned} r_e &= \sqrt{(\bar{x} - D_e)^2 + \bar{y}^2 + \bar{z}^2} \\ &= \sqrt{(-187699.3 - 4674)^2 + (333201.5)^2} \approx 384747.6136 \\ r_m &= \sqrt{(\bar{x} + D_m)^2 + \bar{y}^2 + \bar{z}^2} \\ &= \sqrt{(-187699.3 + 380073)^2 + (333201.5)^2} \approx 384747.8163 \end{aligned}$$

$$\Rightarrow \frac{-\mu_e(\bar{x} - D_e)}{r_e^3} = \frac{-398601(-187699.3 - 4674)}{384747.6136^3} \approx +1.3463 \times 10^{-6}$$

$$-\frac{\mu_m(\bar{x} + D_m)}{r_m^3} = \frac{-4887(-187699.3 + 380073)}{384747.8163^3} \approx -1.6507 \times 10^{-8}$$

$$(d): 1.3628 \times 10^{-6}$$

$$(b): -\omega_0^2 \bar{y} = -\frac{\mu_e \bar{y}}{r_e^3} - \frac{\mu_m \bar{y}}{r_m^3} + \frac{F_y}{m}$$

$$\text{Left side: } -\omega_0^2 \bar{y} = -(2.66199 \times 10^{-6})^2 (333201.5) \approx -2.3611 \times 10^{-6}$$

$$\text{Right side: } -\frac{\mu_e \bar{y}}{r_e^3} - \frac{\mu_m \bar{y}}{r_m^3}$$

$$= -\frac{398601 \times 333201.5}{384747.6136^3} - \frac{4887 \times 333201.5}{384747.8163^3}$$

$$\approx -2.3319 \times 10^{-6} - 2.8590 \times 10^{-8} = -2.3605 \times 10^{-6}$$

(c) Both sides are zero.

Since (a), (b), (c) hold for L_4 , then L_4 is equilibrium point.

b. Define $x = \bar{x} + \delta x$, $y = \bar{y} + \delta y$, and $z = \bar{z} + \delta z = 0 + \delta z$.

$$\begin{aligned} (a) \text{ Left Side: } & (\ddot{x} + \delta \ddot{x}) - 2\omega_0(\dot{y} + \delta \dot{y}) - \omega_0^2(\bar{x} + \delta x) \\ &= \delta \ddot{x} - 2\omega_0 \delta \dot{y} - \omega_0^2 \bar{x} - \omega_0^2 \delta x \end{aligned}$$

$$\begin{aligned} \text{Right Side: } & \frac{\mu_e(\bar{x} + \delta x - D_e)}{r_e^3} + \frac{\mu_m(\bar{x} + \delta x + D_m)}{r_m^3} + \frac{F_x}{m} \\ &= \frac{\mu_e(\bar{x} - D_e)}{r_e^3} + \frac{\mu_m(\bar{x} + D_m)}{r_m^3} + \frac{F_x}{m} + \frac{\mu_e \delta x}{r_e^3} + \frac{\mu_m \delta x}{r_m^3} \\ &= \left[\frac{\mu_e(\bar{x} - D_e)}{r_e^3} + \frac{\mu_m(\bar{x} + D_m)}{r_m^3} + \frac{F_x}{m} \right] + \delta x \left[\frac{\mu_e}{r_e^3} + \frac{\mu_m}{r_m^3} \right] \\ &= \left[\frac{\mu_e(\bar{x} - D_e)}{r_e^3} + \frac{\mu_m(\bar{x} + D_m)}{r_m^3} + \frac{F_x}{m} \right] + \delta x \left[\frac{\mu_e}{r_e^3} + \frac{\mu_m}{r_m^3} \right] \end{aligned}$$

where $\begin{cases} C_1 = (\bar{x} - D_e) / \sqrt{(\bar{x} - D_e)^2 + \bar{y}^2} = (\bar{x} - D_e) / R_e \\ C_2 = \bar{y} / \sqrt{(\bar{x} - D_e)^2 + \bar{y}^2} = \bar{y} / R_e \end{cases}$

$$\begin{aligned} r_m^{-3} &= \sqrt{(\bar{x} + \Delta x + D_m)^2 + (\bar{y} + \Delta y)^2 + \Delta z^2}^{-3/2} \\ &= \sqrt{(\bar{x} + D_m)^2 + \bar{y}^2 + 2\Delta x(\bar{x} + D_m) + 2\Delta y\bar{y}}^{-3/2} \\ &= \sqrt{(\bar{x} + D_m)^2 + \bar{y}^2}^{-3/2} \{1 + 2\Delta x C_3 + 2\Delta y C_4\}^{-3/2} \\ &= \sqrt{(\bar{x} + D_m)^2 + \bar{y}^2}^{-3/2} \{1 - 3C_3\Delta x - 3C_4\Delta y\} \end{aligned}$$

where $\begin{cases} C_3 = (\bar{x} + D_m) / \sqrt{(\bar{x} + D_m)^2 + \bar{y}^2} = (\bar{x} + D_m) / R_m \\ C_4 = \bar{y} / \sqrt{(\bar{x} + D_m)^2 + \bar{y}^2} = \bar{y} / R_m \end{cases}$

$$\Rightarrow - \frac{\mu_e (\bar{x} + \Delta x - D_e)}{r_e^3} - \frac{\mu_m (\bar{x} + \Delta x + D_m)}{r_m^3}$$

$$= - \frac{\mu_e}{R_e^3} (\bar{x} - D_e + \Delta x) (1 - 3\Delta x C_1 - 3\Delta y C_2) - \frac{\mu_m}{R_m^3} (\bar{x} + D_m + \Delta x) (1 - 3\Delta x C_3 - 3\Delta y C_4)$$

$$= - \frac{\mu_e}{R_e^3} (\bar{x} - D_e) - \frac{\mu_m}{R_m^3} (\bar{x} + D_m) - \frac{\mu_e}{R_e^3} \Delta x - \frac{\mu_m}{R_m^3} \Delta x$$

$$+ 3 \frac{\mu_e}{R_e^3} (\bar{x} - D_e) \left(\frac{\bar{x} - D_e}{R_e} \Delta x + \frac{\bar{y}}{R_e} \Delta y \right) + 3 \frac{\mu_m}{R_m^3} (\bar{x} + D_m) \left(\frac{\bar{x} + D_m}{R_m} \Delta x + \frac{\bar{y}}{R_m} \Delta y \right)$$

$$\begin{aligned} \Rightarrow \ddot{\Delta x} - 2\omega_0 \dot{\Delta y} - \omega_0^2 \Delta x &= \frac{\mu_e}{R_e^3} \Delta x \left(\frac{3(\bar{x} - D_e)^2}{R_e} - 1 \right) + \frac{\mu_m}{R_m^3} \Delta x \left(\frac{3(\bar{x} + D_m)^2}{R_m} - 1 \right) \\ &\quad + 3 \left(\frac{\mu_e}{R_e^4} \bar{y} (\bar{x} - D_e) + \frac{\mu_m}{R_m^4} \bar{y} (\bar{x} + D_m) \right) \Delta y + \frac{F_x}{m} \end{aligned}$$

$$\Rightarrow \ddot{\Delta x} - 2\omega_0 \dot{\Delta y} - \omega_0^2 \Delta x = (284991.0000 + 3162.17633) \omega_0^2 \Delta x - 4875710 \omega_0^2 \Delta y + F_x / m$$

$$= 288153.17633 \omega_0^2 \Delta x - 4875710 \omega_0^2 \Delta y + F_x / m$$

$$\therefore \ddot{\Delta x} - 2\omega_0 \dot{\Delta y} - 288153.17633 \omega_0^2 \Delta x + 4875710 \omega_0^2 \Delta y = F_x / m$$

(b) Left side: $\ddot{\Delta y} + 2\omega_0 \dot{\Delta x} - \omega_0^2 \bar{y} - \omega_0^2 \Delta y$

Right side: $-\frac{\mu_e}{R_e^3} (\bar{y} + \Delta y) (1 - 3\Delta x C_1 - 3\Delta y C_2) - \frac{\mu_m}{R_m^3} (\bar{y} + \Delta y) (1 - 3\Delta x C_3 - 3\Delta y C_4)$

$$\begin{aligned} &= - \frac{\mu_e}{R_e^3} \bar{y} - \frac{\mu_m}{R_m^3} \bar{y} - \frac{\mu_e}{R_e^3} \Delta y - \frac{\mu_m}{R_m^3} \Delta y + 3\bar{y} \left(\frac{\bar{x} + D_m}{R_m} \Delta x + \frac{\bar{y}}{R_m} \Delta y \right) \frac{\mu_m}{R_m^3} \\ &\quad + 3\bar{y} \left(\frac{\bar{x} - D_e}{R_e} \Delta x + \frac{\bar{y}}{R_e} \Delta y \right) \frac{\mu_e}{R_e^3} \end{aligned}$$

$$\Rightarrow \ddot{\Delta y} + 2\omega_0 \dot{\Delta x} - \omega_0^2 \Delta y = 3\bar{y} \Delta x \left(\frac{\mu_e (\bar{x} - D_e)}{R_e^4} + \frac{\mu_m (\bar{x} + D_m)}{R_m^4} \right)$$

$$+ \Delta y \left(\frac{3\bar{y}^2}{R_e} - 1 \right) \frac{\mu_e}{R_e^3} + \Delta y \left(\frac{3\bar{y}^2}{R_m} - 1 \right) \frac{\mu_m}{R_m^3} + \frac{F_y}{m}$$

$$= -487569.6068 \omega_0^2 \Delta x + F_y / m$$

$$+ (854481.0479 + 10482.37021) \omega_0^2 \Delta y = F_y / m$$

$$\therefore \ddot{\Delta y} + 2\omega_0 \dot{\Delta x} + 487569.6068 \omega_0^2 \Delta x - 865464.4181 \Delta y = F_y / m$$

$$(c) -\frac{\mu_e \Delta z}{r_e^3} - \frac{\mu_m \Delta z}{r_m^3} = -\frac{\mu_e}{R_e^3} \Delta z (1 - 3C_1 \Delta x - 3C_2 \Delta y) - \frac{\mu_m}{R_m^3} \Delta z (1 - 3C_3 \Delta x - 3C_4 \Delta y)$$

$$= -\frac{\mu_e}{R_e^3} \Delta z - \frac{\mu_m}{R_m^3} \Delta z$$

$$\Rightarrow \ddot{\Delta z} = -\left(\frac{\mu_e}{R_e^3} + \frac{\mu_m}{R_m^3}\right) \Delta z + \frac{F_z}{m}$$

$$\Rightarrow \ddot{\Delta z} + 1.000 \omega_0^2 \Delta z = F_z/m.$$

$$c. \begin{bmatrix} \dot{\Delta z} \\ \ddot{\Delta z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta z \\ \dot{\Delta z} \end{bmatrix} \Rightarrow \left| \lambda I - \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right| = \begin{vmatrix} \lambda & -1 \\ -1 & \lambda \end{vmatrix} = \lambda^2 - 1 = 0.$$

Thus the eigenvalues of out of plane are $\pm j\omega_0$.

d. Since the eigenvalues of out of plane are pure complex number, then the linearized EoM is marginally stable.

Hence we cannot determine the stability of the original nonlinear system.

#3. 해의 안정성

#4. a. Larger, periodic orbit such that is a solution of the nonlinear differential equations of motion.

b. 해의 안정성

AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

October 19, 2016 3:00-5:00 PM

Close Books / Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (40) Consider the motion of a spacecraft subject to the gravity of the Earth modelled as a point mass and its own thrust. Let m be the mass of the spacecraft, $\mu = GM$ be the gravitational parameter of the Earth where G is Newton's gravitational constant and M is the mass of the Earth, $\vec{r}$ be the position vector of the spacecraft, and $\vec{F}$ be the thrust.

a. (5) Derive the equations of motion of the spacecraft in vector form.

b. (5) Let $\vec{r} = xi + yj + zk$, $\vec{F} = F_x i + F_y j + F_z k$ and the state vector $\vec{x} = [x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6]^T = [x \ \dot{x} \ y \ \dot{y} \ z \ \dot{z}]^T$ be defined in the Cartesian coordinate system. Describe the equations of motion in state space form.

c. (15) Let $\vec{r} = r\vec{e}_r + \cancel{v\vec{e}_v} + \cancel{\lambda\vec{e}_\lambda}$ and $\vec{F} = F_r\vec{e}_r + F_v\vec{e}_v + F_\lambda\vec{e}_\lambda$ be defined in the spherical coordinate system with their components in the order of radial, longitudinal, and latitude direction, and $\vec{\omega}$ be the angular velocity vector of the spacecraft. Describe the equations of motion in the spherical coordinate system.

d. (5) Show that the following is an equilibrium solution, i.e., an operating point:

$$\begin{cases} r \equiv R > 0 \\ v = \sqrt{\frac{\mu}{R^3}}t + C, \ C = \text{constant} \\ \lambda \equiv 0 \end{cases}$$

e. (10) Linearize the longitudinal (second) and latitude (third) equations of (c) with respect to the operation point given in (d).

2. (20) Consider a simple double integrator system given in the state space form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

a. (10) If there is no control input, i.e., $u \equiv 0$, discuss the stability of the above system.

b. (10) Show that the above system is controllable. Then, design your own feedback controller in the form of $u = \alpha x_1 + \beta x_2$ such that the control-augmented system becomes asymptotically stable.

3. (20) A spacecraft is instantaneously located at the midway point between the Earth and Moon and the spacecraft has zero velocity vector with respect to the Earth-Moon barycentric coordinate frame. Based on the non-dimensionalized equations of motion for the restricted three-body dynamics, answer the following questions:

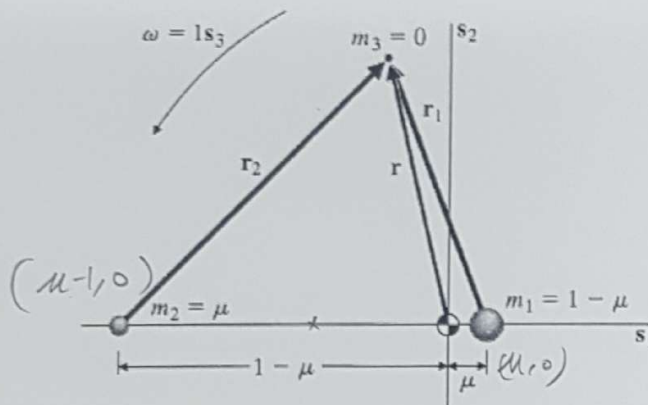


FIGURE 10.1
Geometry of the restricted problem.

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3}$$

$$\ddot{y} + 2\dot{x} - y = -\frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3}$$

$$r_1 = \sqrt{(x-\mu)^2 + y^2 + z^2}, \quad r_2 = \sqrt{(x+1-\mu)^2 + y^2 + z^2}$$

$\mu = 0.0123$ for the Earth-Moon System

- (15) Is the spacecraft in equilibrium at this midway point between the Earth and Moon? If yes, provide your rationale. If not, what is the acceleration vector of the spacecraft with respect to the barycentric coordinate frame?
 - (5) What is the constant (control) acceleration vector that is required to keep the spacecraft in equilibrium at this midway point between the Earth and Moon? Express your control acceleration vector in the barycentric coordinate frame.
4. (20) Consider a two-dimensional linear system

$$\Delta \dot{x} = A \Delta x, \quad \Delta x(0) = \Delta x_0$$

Let $\lambda = \pm j\omega$ and $(\xi, \bar{\xi})$ be the eigenvalues and eigenvectors of the system, respectively. Describe how to find the initial conditions that compose a Halo Orbit with a single frequency of ω .

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 $\mu = G M$

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Newton's Law, $\frac{d^2 \vec{r}}{dt^2} = -\frac{\mu}{r^3} \vec{r} + \frac{\vec{F}}{m}$

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$
 $\dot{\vec{r}} = [\dot{x}, \dot{y}, \dot{z}]^T$ (state)

By $\ddot{x}\hat{i} + \ddot{y}\hat{j} + \ddot{z}\hat{k} = -\frac{\mu}{\sqrt{x^2+y^2+z^2}^3} (x\hat{i} + y\hat{j} + z\hat{k}) + \frac{1}{m}(F_x\hat{i} + F_y\hat{j} + F_z\hat{k})$

$$\begin{cases} \ddot{x} = -\frac{\mu}{(\sqrt{x^2+y^2+z^2})^3} x + \frac{F_x}{m} \\ \ddot{y} = -\frac{\mu}{(\sqrt{x^2+y^2+z^2})^3} y + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu}{(\sqrt{x^2+y^2+z^2})^3} z + \frac{F_z}{m} \end{cases}$$

$$\Rightarrow \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\omega^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\omega^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\omega^2 \end{bmatrix}}_{\equiv A \text{ (system matrix)}} \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{m} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{m} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{m} \end{bmatrix}}_{\equiv B \text{ (control matrix)}} \underbrace{\begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}}_{U \text{ (control)}}$$

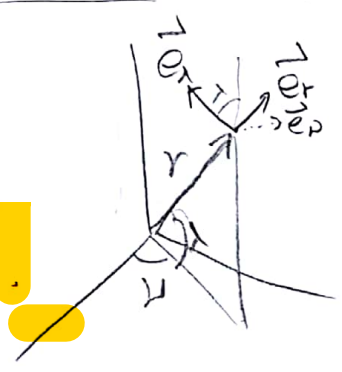
where $\omega^2 = \frac{\mu}{(x^2+y^2+z^2)^{3/2}}$

$\dot{X} = AX + BU$

$\vec{r} = r\vec{e}_r + \nu\vec{e}_\nu + \lambda\vec{e}_\lambda$, $\vec{F} = F_r\vec{e}_r + F_\nu\vec{e}_\nu + F_\lambda\vec{e}_\lambda$

$\dot{\vec{\omega}} = -\dot{\lambda}\vec{e}_r + \dot{\nu}\cos\lambda\vec{e}_\nu + \dot{\nu}\sin\lambda\vec{e}_\lambda$

-1/5



$\dot{\vec{r}} = \dot{r}\vec{e}_r + \dot{\nu}\vec{e}_\nu + \dot{\lambda}\vec{e}_\lambda + r\dot{\vec{e}}_r + \nu\dot{\vec{e}}_\nu + \lambda\dot{\vec{e}}_\lambda$

$= \dot{r}\vec{e}_r + \dot{\nu}\vec{e}_\nu + \dot{\lambda}\vec{e}_\lambda + \vec{\omega} \times (r\vec{e}_r + \nu\vec{e}_\nu + \lambda\vec{e}_\lambda)$ By Coriolis thm.

$= \dot{r}\vec{e}_r + \dot{\nu}\vec{e}_\nu + \dot{\lambda}\vec{e}_\lambda + (-\dot{\lambda}\vec{e}_r + \dot{\nu}\cos\lambda\vec{e}_\nu + \dot{\nu}\sin\lambda\vec{e}_\lambda) \times (r\vec{e}_r + \nu\vec{e}_\nu + \lambda\vec{e}_\lambda)$

$= (\dot{r} + \lambda\dot{\nu}\cos\lambda - \nu\dot{\lambda}\sin\lambda)\vec{e}_r + (\dot{\nu} + \dot{\lambda}\lambda + \dot{\nu}r\sin\lambda)\vec{e}_\nu + (\dot{\lambda} - \dot{\lambda}\nu - r\dot{\nu}\cos\lambda)\vec{e}_\lambda$

$\ddot{\vec{r}} = (\ddot{r} + \lambda\dot{\nu}\cos\lambda + \lambda\ddot{\nu}\cos\lambda - \lambda\dot{\nu}\sin\lambda - \dot{\nu}^2\sin\lambda - \nu\dot{\nu}\sin\lambda - \nu\dot{\lambda}\cos\lambda)\vec{e}_r$

$+ (\ddot{\nu} + \dot{\lambda}^2 + \dot{\nu}^2 + \ddot{r}\sin\lambda + \dot{\nu}\dot{\lambda})\vec{e}_\nu + (\ddot{\lambda} - \dot{\lambda}\nu + \dot{\lambda}\dot{\nu} - \dot{r}\dot{\nu}\cos\lambda - r\dot{\nu}\cos\lambda + r\dot{\nu}\sin\lambda)\vec{e}_\lambda$

$+ \vec{\omega} \times \dot{\vec{r}}$ By Coriolis thm.

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$$\dot{\vec{r}} = (-\dot{\lambda} \vec{e}_r + \dot{\mu} \cos \lambda \vec{e}_\mu + \dot{\mu} \sin \lambda \vec{e}_\lambda) \times$$

$$(\ddot{r} + \lambda \dot{\mu} \cos \lambda - \mu \dot{\lambda} \sin \lambda) \vec{e}_r + (\dot{\mu} + \lambda \dot{\lambda} + \frac{\dot{\mu}^2}{\sin \lambda}) \vec{e}_\mu + (\dot{\lambda} - \lambda \dot{\mu} - \mu \dot{\lambda} \cos \lambda) \vec{e}_\lambda$$

$$= (\dot{\mu} \dot{\lambda} \cos \lambda - \dot{\mu} \dot{\lambda} \cos \lambda - \mu \dot{\lambda}^2 \cos^2 \lambda - \dot{\mu}^2 \sin \lambda - \dot{\mu} \dot{\lambda} \sin \lambda - \dot{\mu}^2 \sin^2 \lambda) \vec{e}_r$$

$$(\dot{\lambda}^2 - \dot{\lambda} \dot{\mu} - \dot{\lambda} \dot{\mu} \cos \lambda + \ddot{r} \dot{\mu} \sin \lambda + \dot{\mu}^2 \lambda \sin \lambda \cos \lambda - \mu \dot{\lambda}^2 \sin^2 \lambda) \vec{e}_\mu +$$

$$(-\dot{\lambda} \dot{\mu} - \dot{\lambda}^2 \lambda - \frac{\dot{\lambda} \dot{\mu}^2}{\sin \lambda} - \ddot{r} \dot{\mu} \cos \lambda - \lambda \dot{\mu}^2 \cos^2 \lambda + \mu \dot{\lambda}^2 \sin \lambda \cos \lambda) \vec{e}_\lambda$$

$$\ddot{\vec{r}} = (\ddot{r} + \frac{\dot{\mu} \dot{\lambda} \cos \lambda}{\sin \lambda} + \frac{\dot{\mu} \dot{\lambda} \cos \lambda}{\sin \lambda} - \dot{\mu} \dot{\lambda} \sin \lambda - \mu \dot{\lambda}^2 \sin \lambda - \mu \dot{\lambda} \sin \lambda - \mu \dot{\lambda} \cos \lambda + \frac{\dot{\mu} \dot{\lambda} \cos \lambda}{\sin \lambda} - \mu \dot{\lambda} \cos \lambda - \mu \dot{\lambda}^2 \cos^2 \lambda - \dot{\mu}^2 \sin \lambda - \dot{\mu} \dot{\lambda} \sin \lambda - \dot{\mu}^2 \sin^2 \lambda) \vec{e}_r$$

$$+ (\ddot{\mu} + \ddot{\lambda} \lambda + \frac{\dot{\lambda}^2}{\sin \lambda} + \ddot{r} \dot{\mu} + \frac{\dot{\mu} \dot{\lambda}^2}{\sin \lambda} - \mu \dot{\lambda}^2 - \ddot{r} \dot{\mu} \cos \lambda + \ddot{r} \dot{\mu} \sin \lambda + \dot{\mu}^2 \lambda \sin \lambda \cos \lambda - \mu \dot{\lambda}^2 \sin^2 \lambda) \vec{e}_\mu$$

$$+ (\ddot{\lambda} - \ddot{\lambda} \mu + \ddot{\lambda} \dot{\mu} - \ddot{r} \dot{\mu} \cos \lambda - \ddot{r} \dot{\mu} \cos \lambda + \ddot{r} \dot{\mu} \sin \lambda - \dot{\mu} \dot{\lambda} - \dot{\lambda}^2 \lambda - \ddot{r} \dot{\mu} \lambda \sin \lambda - \ddot{r} \dot{\mu} \cos \lambda - \lambda \dot{\mu}^2 \cos^2 \lambda + \mu \dot{\lambda}^2 \sin \lambda \cos \lambda) \vec{e}_\lambda$$

$$= \ddot{r} + 2 \frac{\dot{\mu} \dot{\lambda} \cos \lambda}{\sin \lambda} + \frac{\dot{\mu} \dot{\lambda} \cos \lambda}{\sin \lambda} - \ddot{r} \dot{\mu} - \dot{\mu} \dot{\lambda} \sin \lambda - \dot{\mu} \sin \lambda - \mu \dot{\lambda}^2 \sin \lambda - \mu \dot{\lambda} \cos \lambda$$

$$(-\dot{\lambda} \vec{e}_r + \dot{\mu} \cos \lambda \vec{e}_\mu + \dot{\mu} \sin \lambda \vec{e}_\lambda) \times \vec{e}_r$$

$$= -\dot{\mu} \cos \lambda \vec{e}_\lambda + \dot{\mu} \sin \lambda \vec{e}_\mu = \dot{\mu} \sin \lambda \vec{e}_\mu - \dot{\mu} \cos \lambda \vec{e}_\lambda$$

$$(-\dot{\lambda} \vec{e}_r + \dot{\mu} \cos \lambda \vec{e}_\mu + \dot{\mu} \sin \lambda \vec{e}_\lambda) \times \vec{e}_\mu$$

$$= -\dot{\mu} \sin \lambda \vec{e}_r - \dot{\lambda} \vec{e}_\lambda$$

$$(-\dot{\lambda} \vec{e}_r + \dot{\mu} \cos \lambda \vec{e}_\mu + \dot{\mu} \sin \lambda \vec{e}_\lambda) \times \vec{e}_\lambda$$

$$= \dot{\mu} \cos \lambda \vec{e}_r + \dot{\lambda} \vec{e}_\mu$$

$$\ddot{\vec{r}} = (\ddot{r} - \mu \dot{\lambda} \sin \lambda + \dot{\mu} \dot{\lambda} \cos \lambda) \vec{e}_r +$$

$$(\dot{\mu} + \dot{\mu} \dot{r} \sin \lambda + \dot{\lambda} \dot{\lambda}) \vec{e}_\mu +$$

$$(\dot{\lambda} - \dot{\mu} \dot{r} \cos \lambda - \mu \dot{\lambda}) \vec{e}_\lambda$$

$$\ddot{\vec{r}} = (\ddot{r} - \dot{\mu}^2 \sin \lambda - \mu \dot{\lambda}^2 \sin \lambda - \mu \dot{\lambda} \cos \lambda + \dot{\mu} \dot{\lambda} \cos \lambda + \dot{\mu} \dot{\lambda} \cos \lambda - \dot{\mu} \dot{\lambda} \sin \lambda$$

$$- (\dot{\mu} + \dot{\mu} \dot{r} \sin \lambda + \dot{\lambda} \dot{\lambda}) \dot{\mu} \sin \lambda + (\dot{\lambda} - \dot{\mu} \dot{r} \cos \lambda - \mu \dot{\lambda}) \dot{\mu} \cos \lambda$$

$$= \ddot{r} - \dot{\mu}^2 \sin \lambda - \mu \dot{\lambda}^2 \sin \lambda - \mu \dot{\lambda} \cos \lambda + \dot{\mu} \dot{\lambda} \cos \lambda + \dot{\mu} \dot{\lambda} \cos \lambda - \dot{\mu} \dot{\lambda} \sin \lambda$$

$$- \dot{\mu}^2 \sin \lambda - \dot{\mu}^2 \dot{r} \sin \lambda - \dot{\lambda} \dot{\mu} \sin \lambda + \dot{\lambda} \dot{\mu} \cos \lambda - \dot{\mu}^2 \dot{r} \cos \lambda - \mu \dot{\lambda} \cos \lambda$$

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$$= (-\dot{\lambda} \vec{e}_r + \dot{\lambda} \cos \lambda \vec{e}_\theta + \dot{\lambda} \sin \lambda \vec{e}_\lambda) \times$$

$$\{ (\dot{r} + \lambda \dot{\lambda} \cos \lambda - \mu \dot{\lambda} \sin \lambda) \vec{e}_r + (\dot{\lambda} + \lambda \dot{\lambda} + \dot{\lambda} r \sin \lambda) \vec{e}_\theta + (\dot{\lambda} - \dot{\lambda} \mu - \dot{\lambda} r \cos \lambda) \vec{e}_\lambda \}$$

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$$= (\ddot{\lambda} \cos \lambda - \dot{\lambda} \dot{\lambda} \cos \lambda - \dot{\lambda}^2 r \cos^2 \lambda - \dot{\lambda}^2 \sin \lambda - \dot{\lambda} \lambda \dot{\lambda} \sin \lambda - \dot{\lambda}^2 r \sin^2 \lambda) \vec{e}_r +$$

$$(\dot{\lambda}^2 - \dot{\lambda}^2 \mu - \dot{\lambda} \dot{\lambda} r \cos \lambda + \dot{\lambda} \dot{\lambda} \sin \lambda + \dot{\lambda}^2 \lambda \cos \lambda \sin \lambda - \mu \dot{\lambda}^2 \sin^2 \lambda) \vec{e}_\theta +$$

$$(-\dot{\lambda} \dot{\lambda} - \dot{\lambda}^2 \lambda - \dot{\lambda} \dot{\lambda} r \sin \lambda - \dot{\lambda} \dot{\lambda} \cos \lambda - \lambda \dot{\lambda}^2 \cos^2 \lambda - \mu \dot{\lambda}^2 \cos \lambda \sin \lambda) \vec{e}_\lambda$$

$$= (\ddot{r} + 2\dot{\lambda} \dot{\lambda} \cos \lambda - \dot{\lambda}^2 r - 2\dot{\lambda}^2 \sin \lambda + \lambda \dot{\lambda} \cos \lambda - \lambda \dot{\lambda} \sin \lambda - \mu \dot{\lambda}^2 \sin \lambda - \mu \dot{\lambda}^2 \cos \lambda - \dot{\lambda} \lambda \dot{\lambda} \cos \lambda - \dot{\lambda} \lambda \dot{\lambda} \sin \lambda) \vec{e}_r +$$

$$(\ddot{\lambda} + \lambda \dot{\lambda} + \dot{\lambda}^2 + \dot{\lambda} \dot{\lambda} \sin \lambda + \dot{\lambda} \dot{\lambda} \cos \lambda + \dot{\lambda} r \cos \lambda + \dot{\lambda}^2 - \dot{\lambda}^2 \mu - \dot{\lambda} \dot{\lambda} r \cos \lambda + \dot{\lambda} \dot{\lambda} \sin \lambda + \dot{\lambda}^2 \lambda \cos \lambda \sin \lambda - \mu \dot{\lambda}^2 \sin^2 \lambda) \vec{e}_\theta +$$

$$(\ddot{\lambda} - \dot{\lambda} \dot{\lambda} + \dot{\lambda} \dot{\lambda} - \dot{\lambda} \dot{\lambda} \cos \lambda - \dot{\lambda} \dot{\lambda} \cos \lambda + \dot{\lambda} \dot{\lambda} \sin \lambda - \dot{\lambda} \dot{\lambda} - \dot{\lambda}^2 \lambda - \dot{\lambda} \dot{\lambda} r \sin \lambda - \dot{\lambda} \dot{\lambda} \cos \lambda - \lambda \dot{\lambda}^2 \cos^2 \lambda - \mu \dot{\lambda}^2 \cos \lambda \sin \lambda) \vec{e}_\lambda$$

Newton's Law. $\frac{\mu}{r} = -\frac{\mu}{r^3} \vec{r} + \frac{\vec{F}}{m}$

$$\therefore = \ddot{r} - 2\dot{\lambda}^2 \sin \lambda - \dot{\lambda} r + 2\dot{\lambda} \dot{\lambda} \cos \lambda - \lambda \dot{\lambda} \dot{\lambda} (\sin \lambda + \cos \lambda) - \mu \dot{\lambda}^2 \sin \lambda - \mu \dot{\lambda}^2 \cos \lambda + \dot{\lambda} \dot{\lambda} \cos \lambda - \dot{\lambda} \dot{\lambda} \sin \lambda$$

$$= -\frac{\mu}{r^2} + \frac{F_r}{m} \quad \left[\begin{array}{l} \text{e. } \ddot{r} - 2\omega R \dot{\lambda} - 3r \dot{\omega}^2 = F_r/m, \quad R=r. \\ R \ddot{\lambda} + 2\omega \dot{r} = F_\theta/m, \quad \text{where } \omega = \sqrt{\frac{\mu}{R^3}} \\ R \ddot{\lambda} + \omega^2 r \lambda = F_\lambda/m. \end{array} \right]$$

$$\therefore \ddot{r} + 2\dot{\lambda} \dot{\lambda} \sin \lambda + 2\dot{\lambda}^2 + \lambda \dot{\lambda} + \dot{\lambda} \dot{\lambda} \sin \lambda - \mu \dot{\lambda}^2 - \dot{\lambda} \dot{\lambda} \cos \lambda + \dot{\lambda}^2 \lambda \sin \lambda \cos \lambda - \mu \dot{\lambda}^2 \sin^2 \lambda + \dot{\lambda} r \cos \lambda = F_r/m$$

$$\therefore \ddot{\lambda} - \dot{\lambda} \dot{\lambda} + \dot{\lambda} \dot{\lambda} - \dot{\lambda} \dot{\lambda} \cos \lambda - \dot{\lambda} \dot{\lambda} \cos \lambda + \dot{\lambda} \dot{\lambda} \sin \lambda - \dot{\lambda} \dot{\lambda} - \dot{\lambda}^2 \lambda - \dot{\lambda} \dot{\lambda} \sin \lambda - \dot{\lambda} \dot{\lambda} \cos \lambda - \lambda \dot{\lambda}^2 \cos^2 \lambda + \mu \dot{\lambda}^2 \sin \lambda \cos \lambda = F_\lambda/m$$

$r=R, \mu = \sqrt{\frac{\mu}{R^3}} t + C, \lambda = 0 = \dot{r} = \ddot{r} = \dot{\lambda} = \ddot{\lambda} = \dot{\lambda} = \ddot{\lambda}$

or: $-\sqrt{\frac{\mu}{R^3}} R - \sqrt{\frac{\mu}{R^3}} (\sqrt{\frac{\mu}{R^3}} t + C) = -\frac{\mu}{R^3}$

~~10~~

$$X \equiv \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \dot{X} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u.$$

$$\dot{X} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} X \equiv A X.$$

$$\Rightarrow \det(\lambda I - A) = \begin{vmatrix} \lambda & -1 \\ 0 & \lambda \end{vmatrix} = \lambda^2.$$

$$\Rightarrow \text{eigenvalue} : \lambda = 0, 0$$

Since both eigenvalues are 0, then the system is not stable.

한계 시스템이 아닌 시스템.

e^{At} 을 구해보면,

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$$e^{At} = I + At + \frac{1}{2}(At)^2 + \dots$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & t \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

$$X(t) = e^{At} X(0) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1(t) = x_1(0) + x_2(0)t \\ x_2(t) = x_2(0) \end{cases}$$

시스템 stable 하지 않다. (t가 커지면 x_1 이 ∞ 로 간다.)

System matrix : $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, control matrix $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$$\text{rank} \begin{bmatrix} B & AB \end{bmatrix} = \text{rank} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 2.$$

Variable x_2 rank $\begin{bmatrix} B & AB \end{bmatrix}$ 가 같으므로 controllable 하다.

$$u = \alpha x_1 + \beta x_2 = [\alpha \ \beta] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \Lambda X \text{ 라 하면,}$$

$$\dot{X} = AX + Bu = AX + B\Lambda X = (A + B\Lambda)X$$

$$= \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \alpha & \beta \end{bmatrix} \right) X = \begin{bmatrix} 0 & 1 \\ \alpha & \beta \end{bmatrix} X \equiv CX.$$

$$\det(\lambda I - C) = \begin{vmatrix} \lambda & -1 \\ -\alpha & \lambda - \beta \end{vmatrix} = \lambda(\lambda - \beta) - \alpha = \lambda^2 - \beta\lambda - \alpha.$$

$$\Rightarrow \text{eigenvalues } \lambda = \frac{\beta \pm \sqrt{\beta^2 + 4\alpha}}{2}$$

Now consider asymptotically stability.

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Definition is, $\lim_{t \rightarrow \infty} x(t) = x_e$ where x_e is equilibrium point.

$\dot{x} = Cx \Rightarrow x(t) = \exp(Ct) x_0$. where x_0 is initial state.

$$\begin{bmatrix} 0 & 0 \\ \alpha & \beta \end{bmatrix} \begin{bmatrix} 0 & 0 \\ \alpha & \beta \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \alpha\beta & \beta^2 \end{bmatrix} = C^2$$

$$\begin{bmatrix} 0 & 0 \\ \alpha & \beta \end{bmatrix} \begin{bmatrix} 0 & 0 \\ \alpha\beta & \beta^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \alpha\beta^2 & \beta^3 \end{bmatrix} = C^3$$

$$C^n = \begin{bmatrix} 0 & 0 \\ \alpha\beta^{n-1} & \beta^n \end{bmatrix}$$

$$\Rightarrow \exp(Ct) = I + \sum_{n=1}^{\infty} \frac{1}{n!} C^n t^n = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \sum_{n=1}^{\infty} \frac{1}{n!} \begin{bmatrix} 0 & 0 \\ \alpha\beta^{n-1} & \beta^n \end{bmatrix} t^n$$

$$= \begin{bmatrix} 1 & 0 \\ \frac{\alpha}{\beta} \sum_{n=0}^{\infty} \frac{1}{n!} (\beta t)^n - \frac{\alpha}{\beta} & \sum_{n=0}^{\infty} \frac{1}{n!} (\beta t)^n \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ \frac{\alpha}{\beta} (e^{\beta t} - 1) & e^{\beta t} \end{bmatrix}$$

$$\Rightarrow x(t) = \begin{bmatrix} 1 & 0 \\ \frac{\alpha}{\beta} (e^{\beta t} - 1) & e^{\beta t} \end{bmatrix} x_0$$

hence if $\beta \leq 0$, the control - augmented system is asymptotically stable.

이제 controller 설계해야 해서 두자 명시해야 한다.

the position of the spacecraft $\vec{r} = (u-0.5)\hat{i}$

If the spacecraft is in equilibrium, then EoM for restricted 3 body problem hold with zero velocity and zero acceleration.

$$\Rightarrow x = u - 0.5, \quad y = z = 0 = \dot{x} = \dot{y} = \dot{z} = \ddot{x} = \ddot{y} = \ddot{z}$$

$$\Rightarrow \ddot{x} - 2\ddot{y} - x = -\frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3}$$

$$\Rightarrow -x = -\frac{(1-u)(-0.5)}{0.5^3} - \frac{u(0.5)}{0.5^3} = \frac{1}{0.5^2}(1-u-u)$$

$$\Rightarrow -u + 0.5 = \frac{1}{0.5^2}(1-2u)$$

Left side : $-u + 0.5 = 0.5 - 0.0123 = 0.4877$

Right side : $\frac{1}{0.5^2}(1-2u) = \frac{1}{0.5^2}(1-2 \times 0.0123) \approx 3.9016$

Since 0.4877 and 3.9016 are totally different, the spacecraft is not in equilibrium even the other 2 eqs hold.

($\because$) both sides are zero. $\checkmark$

Thus, the acceleration $\vec{a}$ is,

$$\ddot{x} = 2\ddot{y} + x = \frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3}$$

$$= -0.4877 + 3.9016 = 3.4139 \text{ km/s}^2$$

$$\Rightarrow \vec{a} = 3.4139 \text{ km/s}^2$$

To keep the spacecraft in equilibrium, the acceleration should be zero.

$$\Rightarrow \ddot{x} - 2\ddot{y} - x = -\frac{(1-u)(x-u)}{r_1^3} - \frac{u(x+1-u)}{r_2^3} + \frac{F_x}{m}$$

$$\Rightarrow -x + \frac{(1-u)(x-u)}{r_1^3} + \frac{u(x+1-u)}{r_2^3} = \frac{F_x}{m}$$

$$\Rightarrow \frac{F_x}{m} = -3.4139 \text{ km/s}^2$$

$$\Rightarrow \vec{F} = -3.4139 \hat{i} \text{ km/s}^2$$

이제 이 값이 0이 되도록
우변을 0으로 만들어야 한다

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$$\begin{aligned} \dot{\Delta x} &= A \Delta x \Rightarrow \Delta x(t) = e^{At} \Delta x(0) = e^{At} \Delta x_0. \\ &= a_1 \xi e^{j\omega t} + a_2 \bar{\xi} e^{-j\omega t} \\ &= a_1 \xi (\cos \omega t + j \sin \omega t) + a_2 \bar{\xi} (\cos \omega t - j \sin \omega t) \\ &= \underline{(a_1 \xi + a_2 \bar{\xi}) \cos \omega t + j(a_1 \xi - a_2 \bar{\xi}) \sin \omega t}. \end{aligned}$$

Since $\Delta x(0) = \Delta x_0$,

$$\Delta x(0) = a_1 \xi + a_2 \bar{\xi} \quad \& \quad \Delta \dot{x}(0) = (a_1 \xi - a_2 \bar{\xi}) \omega j$$

If $a_1 = a_2$, $\Delta x(0) = 2a_1 \operatorname{Re} \xi$

If $a_1 = -a_2$, $\Delta x(0) = 2a_1 \operatorname{Im} \xi$.

So, the initial condition $\Delta x_0 \in \operatorname{span}\{\operatorname{Re} \xi, \operatorname{Im} \xi\}$ composes a Halo orbit with a single frequency of ω .